Markups on Drop-Downs: Prominence in Pharmaceutical Markets

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Abstract

We study the effect of product prominence in consumer search on demand and equilibrium prices using data from Danish pharmaceutical markets. Variation in prominence comes from alphabetical ordering in physician IT systems. We find that prescriptions, market shares, prices, and revenue all decrease in alphabetical rank. We estimate a structural ordered search model of prescriptions and purchases, and a supply side model of the price equilibrium. Our counterfactuals reveal that sorting products by price instead of name would reduce expenditures by 7.7% in equilibrium, against 2.1% when prices are held fixed. (JEL: D83, L13, D12)

Keywords: Ordered search, pharmaceuticals, market power, prominence.

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1. Introduction

In many retail markets, some products are easier to find than others. This can be a source of market power for firms because prominence reduces the cost of considering their products relative to competing ones. In pharmaceutical markets, prominence may be particularly important. Products are complex to understand for laymen (Bronnenberg et al., 2015) and the search effort involves a multitude of agents, including prescribers and pharmacists, opening a possibility of misaligned incentives and inefficiencies. The economic impact of prominence depends on how it is allocated. While a choice architecture that arbitrarily favors some option can lead to exploitation, an architecture that directs consumers to cheap products becomes an instrument of competition.

The role of prominence and defaults for choice is well documented, in online markets (Dinerstein et al., 2018; Ursu, 2018; Agarwal et al., 2011) and in markets where products are difficult to compare (Madrian and Shea, 2001; Abaluck and Gruber, 2011; Agarwal et al., 2017). Theory is ambiguous about the impact on price: prominence can raise them (Arbatskaya, 2007) or lower them (Armstrong et al., 2009), depending on preferences and search costs. The empirical question of how prominence affects equilibrium prices is harder, because prominence is usually a firm's or a platform's choice.

We study the role of prominence for equilibrium prices and market shares in pharmaceutical markets using a dataset covering all transactions and prices in Danish pharmacies between 2005 and 2017. Prominence arises in three distinct places. First, prescribing physicians use a search engine that lists therapeutically equivalent products alphabetically. The list does not display prices: obtaining a price requires clicking into an individual product, introducing an arbitrary hassle cost of browsing down the alphabet which firms cannot influence. [1] Second, the prescription makes the selected product prominent to the patient. Third, generic substitution obliges the pharmacist to recommend the cheapest available product. The first is our source of exogenous variation, the second transmits it into final demand, and the third — already allocated by price — creates firms' incentive to undercut.

Our setting allows us to overcome two concerns that typically complicate the study of prominence in market settings. First, positioning is often either a choice or a product in itself, and thus an endogenous outcome (as in e.g. Jerath et al., 2011; McDevitt, 2014). Second, in experiments that randomize prominence among consumers, one can estimate the effect on demand (Dinerstein et al., 2018; Ursu, 2018), but not on prices because firms set the same prices for the treatment and control groups. In our setting, alphabetical rank varies across markets and over time and we can treat it as an exogenous shifter of prominence, independent of firm characteristics and predetermined with respect to price.

We find that alphabetical rank affects which drug brand physicians put on the prescription. In our preferred regression specification with product-level fixed effects, the prescription share decreases 5.2%-points per alphabetical rank in duopoly markets. This passes through to consumer purchase so that market share decreases 2.3%-points

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1. We discuss exogeneity in detail in Section 4. We include company and product fixed effects throughout, so variation in rank is driven by factors outside the firm's own control, principally the entry and exit of other products. Firms are moreover institutionally constrained in their ability to change name, as discussed in Section 2.1. That hassle costs of this magnitude materially affect prescribing is documented in Alpert et al. (2024).

in alphabetical rank, and prices decrease 5.3% per rank in duopoly markets. The estimated effect on revenue is 1.8%. In more competitive markets, the effects are numerically smaller but still negative. By including product fixed effects, our results are robust to any time-constant firm-level unobservables that might be correlated with name choice.

While the reduced form results tell us the effect of prominence on prices under the given information structure, we need a model to quantify the relative importance of preferences and information frictions, and to evaluate counterfactual designs of the IT system.

We build a structural econometric model of physician prescribing and consumer purchase. We incorporate this in a pricing game where we estimate firm cost and a payoff dispersion parameter.

On the demand side, physicians exert costly effort to browse through the list of available brands, trading off search costs against expected savings and consumer-product match value. The consumer takes the prescription to the pharmacy where the pharmacist by law is required to recommend the cheapest available product. The consumer then decides whether to sequentially search for alternative products. Thus, the alphabetically first-ranked product is prominent to the prescribing physician, whereas the prescribed and the cheapest products are prominent to the consumer.

On the supply side we model firms as engaging in a pricing game of incomplete information in which each firm receives private payoff shocks. We estimate the dispersion of these shocks jointly with marginal cost by picking the estimates that minimize the distance between predicted and observed price distributions. Our approach is justified by the fact that generic substitution changes the nature of price competition in a way that makes traditional inversion of markups inapplicable.

We document that the demand is discontinuous at the lowest price as undercutting firms get a demand boost — similar to clearinghouse models of price dispersion (Varian, 1980). Firms therefore cannot settle on a single price, since any candidate price invites a small undercut. Instead, the equilibrium object is a distribution of prices. Indeed, relying on FOC's to back out costs based on markups will not work, as it is possible for firms to profitably deviate from such equilibrium by undercutting the price leader.[2]

Estimation of the demand side is made tractable using recent methodological breakthroughs in the estimation of ordered search models. Armstrong (2017) and Choi et al. (2018) showed how to recast "Pandora's Rule" (due to Weitzman, 1979) as a static discrete choice problem, and Moraga-González et al. (2023) derives a search cost distribution implying a closed logit form for the choice probabilities.

The estimated demand model shows that physicians are not ignorant of the consequences for the consumer, but neither are they perfect search agents. Search effort responds to what is at stake for the patient - physicians search more for poorer consumers. On the other hand, physicians respond less to variance in consumer expenditures than the consumer does. A 1% rise in expected patient expense lowers prescribing probability by 0.23%, versus 1.24% for a 1% price rise in consumer purchase probability.

2. We show this formally in Appendix C.

We use the model to compare three interventions in the prescription decision. First, removing the alphabetical ranking friction accomplishes little, and may even drive demand towards more expensive options when branded ranks last. Second, ranking products by price rather than alphabetically reduces expenditures by 7.7%. Holding prices fixed, the same reform saves about 2.1%, so roughly three quarters of the total savings can be attributed to price responses. Lastly, mandating that the cheapest product is prescribed saves 25.3%, but restricts clinical discretion in a way that remains contested.

Price ranking is therefore an unusually cheap policy instrument: it is a change to a sort order, restricts no one's discretion, and requires no new regulation. It works precisely because it does not remove the friction, but redirects the loyal demand the friction creates toward whichever firm is cheapest. Electronic prescribing, generic substitution and reference pricing are common across European health systems, so the design margin is not specific to Denmark.

The broader lesson concerns market architecture rather than behavioral frictions. Consumers make systematic mistakes in markets for health insurance (Abaluck and Gruber, 2011; Bhargava et al., 2017), retirement savings (Madrian and Shea, 2001; Benartzi and Thaler, 2007) and mortgages (Agarwal et al., 2017), and a large literature asks whether and how the resulting frictions should be removed.[3] We argue that the more consequential question is to whom the architecture delivers the loyal demand it

3. See e.g. Abaluck and Gruber (2016); Ericson and Sydnor (2017); Ketcham et al. (2019).

creates. Formularies, platform rankings, default investment options and procurement rules all face this choice.[4]

Our paper contributes first to the literature on pharmaceutical demand and competition. Previous work has shown that demand for branded and generic pharmaceuticals is shaped by information and expertise (Bronnenberg et al., 2015), switching costs and experience (Janssen, 2023), advertising (Sinkinson and Starc, 2019), and generic substitution (Janssen and Granlund, 2023). This literature has largely studied pricing as it is shaped by regulation, bargaining and market structure, taking as given the rules that allocate demand across therapeutically equivalent products. We study those rules directly: generic substitution creates a discontinuous advantage for the lowest-priced product, and we develop a framework for analyzing pricing in its presence.[5]

Our paper also contributes to a literature on health information technology and physician decisions. Previous work shows that access to drug information can change prescribing behaviour (Arrow et al., 2020), that machine-learning tools can complement physician expertise (Ribers and Ullrich, 2024, 2023), but also that the hassle costs imposed by some IT systems can materially affect prescribing (Alpert et al., 2024; Böckerman et al., 2025). Consistent with the importance of

4. Empirical work has documented important effects of search frictions on prices in mutual funds (Hortaçsu and Syverson, 2004), social-security plans (Hastings et al., 2017), credit cards (Galenianos and Gavazza, 2018), mortgages (Allen et al., 2019), and online markets (Ellison and Ellison, 2018).

5. For a survey of structural work on prescription pharmaceutical markets, see Ching et al. (2019). Related work on pharmaceutical pricing and regulation includes Dubois and Lasio (2018) and Kortelainen et al. (2023); the former confronts a related methodological difficulty, recovering marginal costs from unconstrained markets where the standard inversion is unavailable.

software design for this margin, Patel et al. (2014) show that changing the default in an electronic prescribing system substantially increases generic prescribing. Our contribution is to show how and why the design of the prescribing interface affects equilibrium prices.

Finally, we contribute to the literature connecting search architecture to equilibrium competition. Closest to our paper, Dinerstein et al. (2018) show that the design of an online search platform steers demand and strengthens sellers' incentives to lower prices, while Brown (2019) and Luco (2019) document equilibrium price responses to information provision. Our setting highlights a distinct role for market architecture: it determines how the valuable demand generated by prominence is allocated. A separate and largely theoretical tradition has long shown that a discrete advantage to being cheapest destroys deterministic pricing and generates equilibrium dispersion (Varian, 1980; Baye and Morgan, 2001). In those models the discontinuity arises from heterogeneity in consumers' information. In ours it is created deliberately by an institution, which is what makes it available as a policy instrument, and what allows us to estimate its consequences.

The rest of the paper is organized as follows. Section 2 describes the data and the institutional setting. Section 3 presents descriptive evidence and Section 4 regression results. Sections 5 and 6 present our structural models for demand and supply respectively, which we use for our counterfactual simulations in Section 7.

2. Data and Institutional Setting

2.1. Institutional Setting

Denmark has a universal single-payer healthcare system which also subsidizes prescription drugs. To contain costs, the government therefore regulates both the supply and demand side which we will describe in turn.

Prices: There are three relevant prices in this setting: the wholesale price, the retail price, and the out-of-pocket price. Wholesale prices are submitted by firms in a centralized mechanism operated by the Danish Medicines Agency (DMA). Every 14 days, firms submit wholesale prices for all their products to the DMA. Throughout, we will refer to the 14-day period as a *period*. The retail price is the wholesale price plus a fixed pharmacy fee, which is set by the government. This fee is a stepwise linear function of the wholesale price, which changes over time, and we recover from the transaction data. Finally, the out-of-pocket price is the retail price minus a government subsidy, which we describe later. The pharmacist earns the pharmacy fee, and the pharmaceutical firm gets the wholesale price, but pays something to the distributor/wholesaler from that, which is not public data. See more details in Appendix B.

Markets: Products are grouped into so-called "substitution groups", which is a set of products having the same substance, strength, dose and similar pack size. [6] All

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6. Pack size is measured in number of Defined Daily Dosages (DDDs). The definition of a substitution group allows for a variation of $\pm 10\%$ in DDD within the group. Products in a substitution group are bioequivalent, but not necessarily chemically identical. In rare cases it is possible that this distinction makes a difference, for instance if consumers are allergic to the coating of one product but not the other.

products that have a submitted price in the system are available for the consumer to purchase at that price, and a consumer is allowed to choose any product in the same substitution group as the product on her prescription.[7]

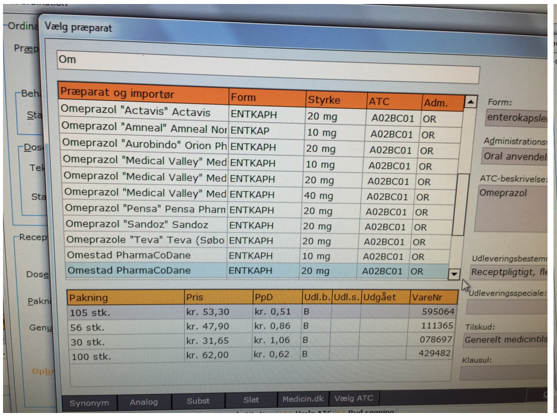
Physicians: To buy a drug, the consumer must first obtain a prescription from a physician. Consumers can freely choose their physician, although changing the physician can be costly. When writing a prescription for the patient, the physician has no monetary incentives over products and must choose one of the available products in the substitution group; it is not possible for the physician to specify that she has no preference for one firm's generic over another in the substitution group. It is however possible for the physician to tick off a box which blocks the pharmacist from substituting the prescribed product with a cheaper alternative. We observe when this is used (4% of all transactions). Physicians make their choice in an IT system, which transfers the prescription electronically to the pharmacy. We note that while patients are free to choose physicians, patients are also always free to substitute away from the brand that a physician puts on the prescription. Therefore, if a consumer is sophisticated enough to search for a physician based on prescription practices, the consumer should be sophisticated enough to undo the prescription at the pharmacy if she disagrees. This makes endogenous patient-physician matching (Dubois and Tunçel, 2021; Huang and Ullrich, 2026) less of an issue for our question.

For the vast majority of consumers, products in a substitution group are for all practical purposes perfect substitutes, although our structural model does not assume this.

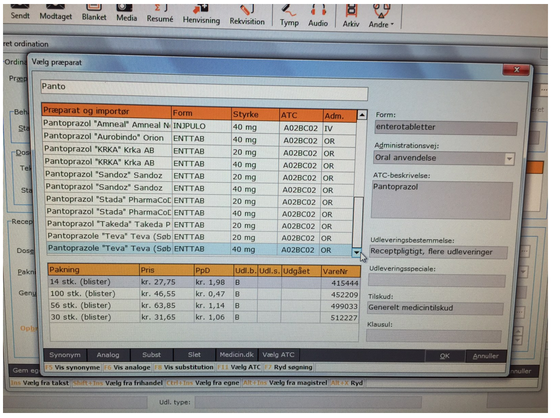
7. Technically, the consumer can also substitute, say, two 25 pill packages for one 50 pill package of the same product, so long as the quantity does not increase.

FIGURE 1. Search engine illustration

(A) Example 1: Omeprazole



(B) Example 2: Pantoprazole



Alphabetical ranking: As mentioned, our empirical strategy relies on variation in prominence generated by the alphabetical ranking of products. The alphabetical ranking arises due to the design of the search tool in the physicians' IT system. When physicians type in a query, e.g. "Omeprazole", packages with a name matching the search term are presented in alphabetical order by the name of the package, as illustrated in Figure 1. In this way, the name of the product becomes important. If physicians want to search for the cheapest product, they have to click on each product to view additional information such as price which is hidden in the default view.[8]

The ability of firms to change the names of their products is limited by regulation. Under Danish law, pharmaceutical product names can take one of two forms: *1)* A *special name* which is to be approved by the government, and must comply with

8. We first became aware of the UI design when an irritated physician explained how she spent time clicking through the list to find a cheap product to put on the prescription. Other physicians we have talked to have no reason to pick one product over another, expecting the pharmacist to help the consumer to find a cheaper alternative.

a number of rules (It must be different from all other international non-proprietary names for instance, must be pronounceable etc.). *2)* A Danish or international non-proprietary drug name (e.g. penicillin) followed by the company name, i.e. `Molecule ("Firm A")`. Typically, the original manufacturer will fall into category 1, while most generics fall into category 2.

Firm types: The most critical distinction is between "generic" and "branded" products. We will define a product as being generic if there are quotes in the name, and otherwise we will refer to it as "branded". This ensures that in the markets we study, the order of the generic firms in the search rank is determined by the company name. Conversely, the branded product will appear either before or after the block of generic products depending on the alphabetical ranking of the proprietary and non-proprietary names of the molecule. [9] A third product type is "parallel imported" products. These are products that have been purchased in another country and imported to Denmark. Parallel imports make up 18% of our product-period observations, although they only account for 2.0% of transactions, and 83.9% of parallel imports are non-generics. Usually, parallel imports are major players prior to loss of exclusivity, but they are also present after.

Generic substitution: The pharmacist's role in the mechanism is to guide demand towards the cheapest product in the market. Pharmacists are legally mandated to

9. For example, ATC N03AX09 has proprietary name Lamictal and non-proprietary name Lamotrigin, whereas ATC C10AA01 has names Zocor and Simvastatin. For those two markets, the branded firm thus comes before (N03AX09) and after (C10AA01) generics, respectively.

recommend that the consumer buys the cheapest product in the substitution group.[10] This is referred to as “generic substitution” and appears in most countries in some version. As a result, when one firm undercuts another, even by a tiny amount, this makes it the recommended product made by all pharmacists. This explains the strong discontinuity in market demand at the minimum price that we will present later. Finally, the authorities refer to the “A” products as the cheapest, and “B” products are any with price within 5% of the cheapest. The rest are designated “C”. The law uses different language for the generic substitution mandate depending on whether the prescribed product is B or C, and the classification appears in the pharmacy IT systems, so we will control for it in our demand model. We cover additional details regarding pharmacy regulations in Appendix B.

Co-insurance: In the end, the consumer makes the final choice and pays for the product. The consumer receives a fraction of the cheapest price in subsidy. This fraction increases in annual expenditure, starting at 0% and eventually reaching 100% for consumers with very high accumulated expenditures (see Appendix Figure B.1 for details). However, a consumer must pay the full price difference between the cheapest product and the chosen product out of her own pocket. This structure of insurance implies that the consumer receives a fixed subsidy regardless of which product she chooses and then pays the price difference down to the cheapest. This allows us to simplify the structural model and abstract from the subsidy. While some

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10. If multiple firms bid the lowest price, the pharmacist may choose which to recommend. In practice, pharmacies often conduct the substitution even if products are less than 5% more expensive than the cheapest product.

consumers have additional market-based health insurance, this insurance does not alter the relative prices.[11] Additional institutional details are in Appendix B.

To summarize, the firm submitting the lowest price in a market – the "winner" – is rewarded in two ways in the mechanism: first, through the prominence resulting from the pharmacist's recommendation, and second, through the subsidy structure, since co-insurance kicks in under the minimum price. The physician's decision to prescribe one product over another in the same market does not affect the consumer apart from making that product the de facto default. The alphabetical ordering affects the physician's prescription probabilities, which in turn affects product prominence to the consumer through the default status.

2.2. Data

Our dataset is merged from a number of sources. Most importantly, we rely on the universe of all transactions of prescription drugs in Danish pharmacies in the period April 2005 to June 2017. Each row in that dataset contains an identifier for the purchasing consumer, the prescribing physician, and product identifiers for the purchased and prescribed product (which may be different). In addition, each row contains product and consumer information such as the price, pack size, form

11. Private health insurers offering supplementary coverage are also active in the Danish market, the largest such provider being "danmark". The most common package in "danmark" covers an additional part of the out-of-pocket payment but, similarly to the public insurance, not the extra price differences from opting out of the cheapest. This means that the price differences between the available products are not affected by insurance. Since we do not have an outside option in our model, discrete choice probabilities will be numerically unaltered. We do not know much about the outside option because we do not observe prescriptions that are not filled.

and the subsidy received. We augment this dataset with all prices from an online source maintained by the Danish Medicines Agency (DMA) who runs pharmaceutical auctions. This gives us information on available products that did not sell in a two-week period and thus did not appear as a transaction. We construct patent expiration dates by combining data on special European patent extension dates (SPCs) from the Danish Patent and Trademark Office with data on molecule marketing approval dates from the DMA. We merge this data with consumer demographics (age, income, gender and education) from Danish population registers. We do not observe whether a product is generic or branded directly. Instead, we define a product as generic if it has quotation marks in its name, which indicates that a product belongs to naming category 2 as described above. We also run robustness checks with a broader definition, where we define a generic as a product that was not present before patent expiration in a market where we observed at least one such.

2.2.1. Sample selection. We will use two different datasets in this paper: first, a market-level dataset with 749,859 product-period observations. We use these for our reduced-form results. Second, a transaction-level dataset, where we choose a random subsample of 967,914 transactions to estimate our structural model of demand and supply.

The market-level sample construction is summarized in Table A.1. Our sample goes from April 1st 2005 to June 19th 2017, which is the last period of data we have. The start date is chosen to be after a reform that drastically changed the pricing system (see Kaiser et al., 2014). We only keep product-periods that satisfy the following criteria: minimum one year after patent expiration; at least one generic product present; product name (in the IT system) must be observed; and between two and eight firms active.

For our transaction-level dataset, Table A.2 shows the sample selection. We start from all transactions of products that satisfy the same criteria as above, which covers 232 million transactions. We first take a random subsample of 3 million transactions. We then restrict attention to periods where at least two generic products were present: since we will treat non-generic products as non-strategic, two is the minimum number required for active competition. This skews the sample towards larger markets and drops 55% of transactions.[12] We then drop transactions where there were missings, prescriptions in different markets, missing demographics (which drops a year), where the physician prohibited substitution, and where subsidy information was missing. This lands us at 967,914 transactions, or 32%.

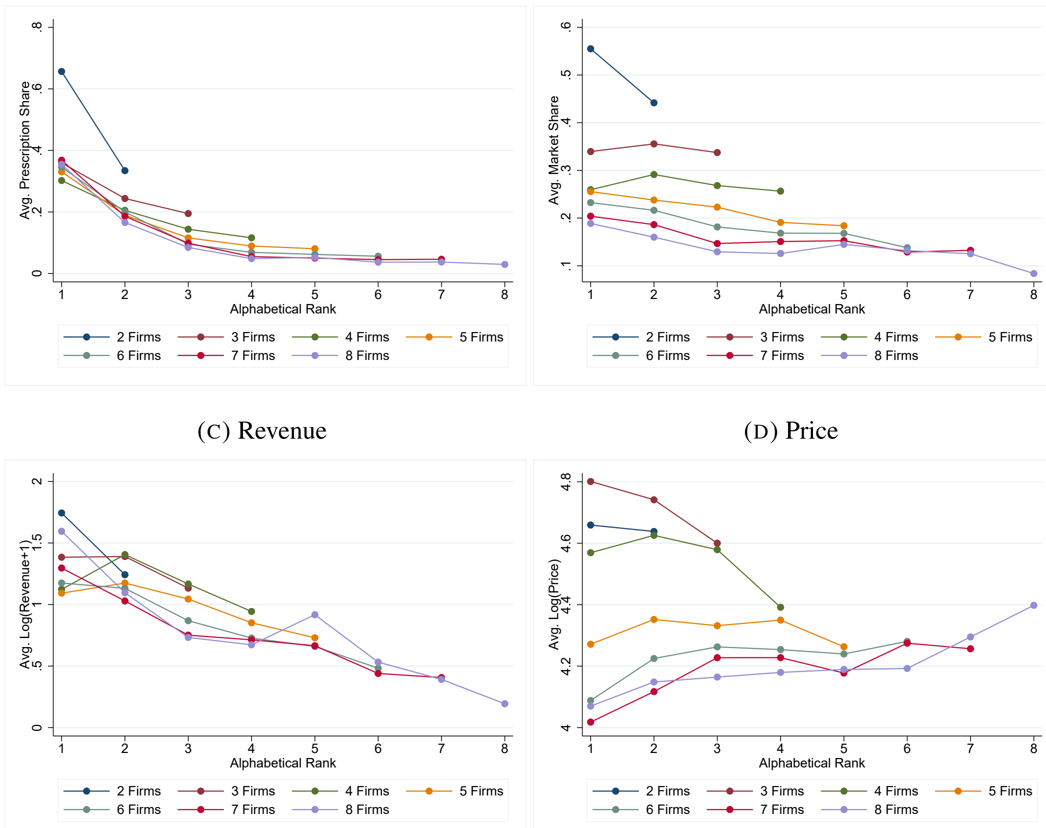
3. Descriptive evidence

Figure 2 shows the relationship between the alphabetical rank of a product on the x-axis and the average or median of four different outcome variables on the y-axis, and as such displays the raw associations in the data to provide an overview.

The reason why the alphabetical rank of a product matters to market outcomes is that it affects physician prescriptions. We illustrate the relation between prescriptions and alphabetic rank for generic pharmaceutical products in Figure 2a. The graph

12. We require at least two generic products. This naturally causes us to drop disproportionately many transactions from smaller markets: we retain only 14.7% of 2-firm markets, but 63.9% of 8-firm markets. This is naturally a skew, but necessary to study active competition among generics.

FIGURE 2. Alphabetical Rank and Market Outcomes for Generics



(A) Prescription share (B) Market share (C) Revenue (D) Price

Note: All figures are constructed using our product-level dataset (i.e. an observation is a product-period) including only generic products. Since 5% of observations have zero revenue we add 1 to revenue (in 1000 DKK) before taking logs.

shows that the earlier in the alphabet a firm is, the more prescriptions it obtains. Furthermore, the effect flattens out from rank 4 and onwards, indicating that the effect of rank is most important for the first couple of clicks down the list in the IT system.

Figures 2b and 2c show that both market share and revenue are declining in alphabetical rank, although as one might expect, the relationship is less stark. This shows that being prominent appears to be an advantage to firms.

Figure 2d shows the average log unit price by the alphabetical rank of the firm. The relation between price and alphabetical rank appears to depend on market structure. For markets with two, three, and four firms the slope is negative, but in less concentrated markets the price-rank gradient flattens.

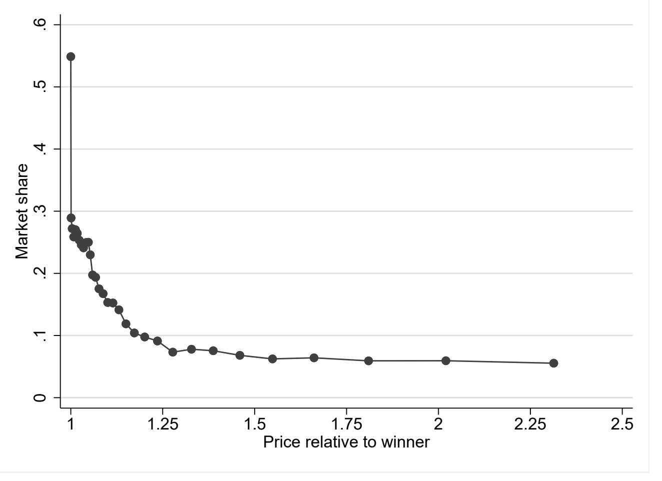
All the panels in Figure 2 are constructed using only observations for generic products, which are the ones that are ranked by the firm name. Original manufacturer products use the proprietary name, which can therefore be either first or last, as explained earlier.[13] Appendix Figure A.1 contains the same panels including all products. As expected, the first and last products look different there. When we later turn to regressions, we can control for whether a product is generic to avoid such compositional changes, and our results are robust to running either on the full sample or only generics, so we have opted to use the most inclusive sample in the remainder of the paper.

3.1. The shape of demand

Next, we turn to the shape of the demand curve, describing how patient purchase depends on price. Figure 3 shows on the x axis the price relative to the winning price in the market, and on the y axis the average market share within bins. When the relative price is 1, it means that the product had the cheapest price in the market during that period, and on average such products received about 55% market share. For the product-periods with a price just a tiny bit above the minimum price, the market share was instead just under 30%. This sharp discontinuity at the minimum price is due to *generic substitution*, whereby the pharmacist is mandated to recommend

13. In a few cases, packages can have non-generic names for other reasons (e.g. parallel imported generics). Appendix Figure A.2 shows the frequency with which a given product is a generic for each rank position.

FIGURE 3. Demand discontinuity at the minimum price



Note: The figure shows binned averages of the market share of products plotted against the price relative to the cheapest product on the x-axis. For example, an x-value of 1.0 indicates that the product was the cheapest, whereas a value of 2.0 indicates a price 100% higher than the minimum price. An observation is a product-period. Note that because the plot includes products across markets with different numbers of firms, and because firms sometimes tie at the cheapest, we should not expect the plot to integrate to one.

that the consumer buy the cheapest available product. This discontinuity in demand eliminates pure strategy equilibria, since firms have a strong incentive to undercut. However, Figure 3 also shows that a non-negligible market share accrues to products with very high prices: products with prices more than 100% over the minimum price managed to still attract a 10% market share on average. These two features together – a discontinuity at the bottom, but a non-zero market share at high prices – make the demand side reminiscent of the "shoppers and loyals" model of Varian (1980).

The same pattern can be seen at the level of individual transactions, and there we can more clearly see the role of the prescription in shaping the upper tail of demand. Table A.7 in the appendix shows that the probability that a consumer buys the prescribed product is 78.2% when the prescribed product is the cheapest in the market ("A" region). This falls to 49.1% if the prescribed product is within 5% of the cheapest ("B" region), and to 24.4% if the prescribed product is more than 5% above

TABLE 1. Variance Decomposition: Within and Between Products

	Mean	Overall σ	Between σ	Within σ
Rank	2.34	1.53	1.53	0.54
Log(Price)	1.22	1.57	1.58	0.35
Log(Revenue)	1.87	1.49	1.17	0.93
Prescription share	0.284	0.27	0.24	0.11
Market share	0.286	0.30	0.20	0.23
Observations				749,859

Note: Computed based on 8,487 unique products over 88.4 price periods (on average). The Overall σ is the usual standard deviation, while the Between σ is the standard deviation of product averages, and the Within σ is the standard deviation within products over time.

the cheapest (“C” region). Substitution at the pharmacy is therefore quantitatively important, but there are still many consumers who end up following the prescription in spite of cheaper options. It is this incomplete pass-through from the pharmacy counter back to the prescription that gives the physician’s choice, and hence the ordering of products in the physician IT system, its influence over firm demand.

3.2. Sources of variation in alphabetical rank

We start by considering how much variation we have in the key variables between products in the cross-section versus over time within products. Table 1 presents variance decompositions of the key variables into the overall, between, and within components. Most crucially, we note that all variables display considerable variation within products over time, which is what our fixed effects specifications will exploit. And for the market share, the within variation is actually greater than the between variation, which makes sense in light of the discontinuity in demand documented in Figure 3.

TABLE 2. Change in Alphabetical Rank from Period t to $t+1$

Change in rank	Frequency	Percent
-2 or lower	823	0.1
-1	14,879	2.0
0	709,603	95.9
+1	13,278	1.8
+2 or higher	1,138	0.2
Total	739,721	100.0

Next, let us turn to our main outcome variable, price, and our main explanatory variable, alphabetical rank. First, we note that being the cheapest product in the market is far from constant: only 72.1% of products that are cheapest in period t are still cheapest in period $t+1$. Conversely, the 15.3% of products that are not cheapest in t become cheapest in $t+1$. So there is considerable churn in the cheapest product. This is consistent with the undercutting motive implied by generic substitution.

Alphabetical rank, however, is far more stable. Table 2 shows the distribution of changes in rank from one period to the next. 95.9% of product-period observations have no change in rank, with 2.1% falling and 2.0% rising in rank, concentrated at one step.

Changes in rank can only come from the entry or exit of competitors that sort earlier in the alphabet. The effect is mechanical: if a product enters at rank r, then all firms previously having rank r or higher are demoted by one position, while the products ranking before r are unaffected.[14] And similarly for exit events. Short of these, a product's rank remains fixed over time in our data.

A natural concern is whether entry is endogenous to rank. Table 3 tabulates all entry and exit events in our sample by the rank at which they occur and the size of the market.[15] Entering products are indeed somewhat early-sorting relative to a uniform benchmark – the mean rank of an entrant is 2.31 against a benchmark of 2.66 – but so are exiting products, whose mean rank is 2.47 against a benchmark of 2.79. The symmetry is close at every market size: in Table 3 the mean entry rank and the mean exit rank never differ by more than 0.2 positions for any J. The tilt toward early positions therefore reflects the composition of the high-turnover segment of the market rather than strategic acquisition of prominent positions. Were early ranks being competed for, entry would be concentrated at rank 1 while exit was not; instead both are.

4. Regression Results

In this section, we estimate the effect of alphabetical rank on prescription share, price, market share and revenue share using fixed effects regressions. The unit of observation

14. The only exception occurs if there are multiple products with identical package names, which can happen. In that case, we have opted for a convention of assigning each of these the same rank. So a market with products A, B, B, and C, would have ranks 1, 2, 2, and 3, with the two B products sharing rank 2, and the last product being one higher, so there is no rank without a product in the middle.

15. An entry event is defined as the first period in which a product is observed in a market, and an exit event as the last; products present in the first (last) period of the sample are not counted as entering (exiting).

TABLE 3. Entry and exit events by alphabetical rank and market size

	Number of firms in market (J)							
	2	3	4	5	6	7	8	Total
Entry Events	648	693	623	571	438	331	219	3,523
Entrant mean rank	1.32	1.60	2.04	2.63	3.01	3.66	4.23	2.33
Exit Events	684	760	764	847	686	487	315	4,543
Exiting product mean rank	1.38	1.73	2.25	2.65	2.93	3.62	4.15	2.49
Uniform benchmark $(J+1)/2$	1.50	2.00	2.50	3.00	3.50	4.00	4.50	–

Note: An observation is an entry or exit event. In Panel A, J is the number of products in the market after entry; in Panel B, J is the number of products before exit. The uniform benchmark $(J+1)/2$ is the mean rank that would obtain if entering (exiting) products were drawn at random from the alphabetical ordering; if we compute the J-weighted benchmark, it is 2.66 for entrants and 2.79 for exiting products. Based on Appendix Table A.6.

TABLE 4. Summary Statistics

	Mean	S.d.	P10	P50	P90
No. Firms	4.20	1.77	2.00	4.00	7.00
Log(Revenue+1)	1.87	1.49	0.09	1.64	3.96
Log(Price)	1.22	1.57	-0.46	0.94	3.19
Price relative to winner	1.86	3.99	1.00	1.04	2.34
Alphabetical Rank	2.34	1.53	1.00	2.00	5.00
Market Share	0.29	0.30	0.00	0.16	0.79
Prescription Share	0.28	0.27	0.02	0.19	0.72
Quantity sold (1000 packages)	0.36	1.28	0.00	0.05	0.78
Years since end of exclusivity	11.00	9.06	2.36	8.30	23.60
No. Generics (def. 1)	2.21	2.07	0.00	2.00	5.00
No. Generics (def. 2)	3.21	1.84	1.00	3.00	6.00
Rank 1 is generic	0.35	0.48	0.00	0.00	1.00
N (product-periods)			749,859		

is a product, j, in a two-week period, t. We cluster standard errors at the market level (substitution gruop). Our estimation sample has 749,859 product-period observations across 1,573 markets. However, in some specifications, there are panel units without enough variation which gets dropped. We present summary statistics for our main regressions sample in Table 4.

4.1. Econometric Specification and Identification

We will present results for four different outcomes, collectively labelled y_{jt}: the logarithm of the unit price, the prescription share, the market share, and the revenue share.[16] We consider variations of the following regression model:

$$y_{jt} = \rho_1 R_{jt} + \rho_2 R_{jt} \times J_{m_j t} + \sum_{k=3}^{8} \delta_k \mathbf{1}(J_{mt} = k) + \mathbf{x}_{jt}\beta + \eta_{jt} + \varepsilon_{jt}, \tag{1}$$

where m_j is the market to which product j belongs, $J_{m_j t}$ is the number of products available in the market in period t (so δ_k are dummies for the number of active firms), $R_{jt} \in \{1,...,J_{m_j t}\}$ is the alphabetical rank , $\mathbf{x}_{jt}$ is a vector of controls, which includes dummies for the number of years since patent expiration. Lastly η_{jt} is short-hand for time and/or product fixed effects. We use different specifications for η_{jt} to use different sources of identifying variation. We allow the effect of rank R_{jt} to depend on the level of competition by including the interaction between rank and number of firms in the market, so the effect of rank is composed of the linear effects, ρ_1,ρ_2. In the following, we discuss identification under various specifications of η_{jt}.

We use fixed effects to address the possibility that firms early in the alphabet are fundamentally different from firms late in the alphabet, something that would bias pooled estimates. Our main concerns are 1) that some firms may game the system by strategically choosing a name that is early in the alphabet and 2) that entry may be endogenous to alphabetic rank and demand factors. We address the first concern by including firm fixed effects f_j (of which there are 72), in all specifications. This

16. The revenue share is defined as the revenue (price times quantity) for product j out of total revenue earned by all firms in the same market and period.

means that if one firm always ranks first whenever it is active, its outcomes will not provide variation the contributes in identifying ρ_1, ρ_2. It will, however, affect the rank of other firms, since it may push competitors further down the list when it enters. In some specifications, we also include market fixed effects, which for example absorbs time constant market-specific demand factors, such as therapeutic area of the drug or the cost of production.

To address potential differences between products early and late in the alphabet we further estimate a specification where we use product fixed effects, η_j. The product fixed effect is identical to a fixed effect for each firm-market pair, since products do not change owning firms in our setting.[17] Therefore, the product fixed effect would in particular absorb any firm or market fixed effects. Moreover, it captures any differences in product characteristics that might exist between (bioequivalent) drugs in a market: color of the coating, shape of the pills or brand perception. In this specification, the variation in alphabetical rank comes from entry or exit of competitors with names positioned earlier in the alphabet. In one market, a competitor may enter that does not affect the rank of firm j (if the entrant is later in the alphabet), whereas another firm may enter that overtakes firm j, reducing its rank. In this way we get separate identification of the alphabetical rank and the number of firms conditional on η_j. Finally, this specification is robust to entry on time-constant unobservables since this is essentially an effect where specific firms (e.g. last in the alphabetical list) only tend to enter specific markets (e.g. ones that have particularly inelastic consumers).

17. If one generic firm acquires the legal right to sell a product from another firm, it would result in a new product ID and the name of the package would have the new owner's firm name.

Lastly, we implement a specification that addresses entry on time-varying market unobservables by considering a specification with market-by-date fixed effects, complemented by firm (f_j) fixed effects, by setting $\eta_{jt} = \eta_{m_j t} + \eta_{f_j}$. Here, there is only variation in alphabetical rank within a market across the different products, so the identifying variation for ρ_1, ρ_2 is instead based on a direct comparison between products of different ranks. Most importantly, this specification controls for time-varying market-specific unobservables. Imagine for example that high-ranking firms are only active when demand is extremely high, and that this high demand causes prices to increase in general (creating endogeneity). That effect would be captured by $\eta_{m_j t}$.

4.2. Results

We present results for the four outcome variables of interest in Table 6. Each panel corresopnds to an outcome. In all specifications, the marginal effect of rank under a given market structure is computed as the sum of the direct effect (ρ_1) and interaction effect with number of products, $\rho_1 + \rho_2 \times J$. To avoid evaluating this expression, Table 5 summarizes all the following tables, showing the results from our preferred specification with product-level fixed effects for all four outcomes. We see that all four outcomes are decreasing in alphabetical rank, but that the slope becomes numerically smaller when more firms are present in the market. In the following, we will go through each outcome separately and compare results across specifications of η_{jt} and discuss statistical significance.

Table 6 Panel A shows results for the effect of alphabetical rank on the share of prescriptions in a market accruing to a specific product. Across specifications we find

TABLE 5. Marginal Rank Effects and Market Structure

	Log Price	Prescription Share	Market Share	Revenue Share
$J = 2$	-0.053	-0.052	-0.023	-0.018
$J = 3$	-0.044	-0.045	-0.022	-0.016
$J = 4$	-0.036	-0.038	-0.020	-0.014
$J = 5$	-0.027	-0.032	-0.019	-0.012
$J = 6$	-0.019	-0.025	-0.018	-0.010
$J = 7$	-0.010	-0.019	-0.016	-0.008
$J = 8$	-0.002	-0.012	-0.015	-0.006

Note: The table shows the effect of alphabetical rank on each respective outcome variable for different levels of competition measured as the number of available products (J). The effects are computed as $\hat{\rho}_1 + \hat{\rho}_2 J$. The marginal effects are computed using our preferred regression specification, column (4) of Table 6, which uses fixed effects at the product and period level: $\eta_{jt} = \eta_j + \eta_t$. The reported quantities are linear transformations of two estimates from 6, where standard errors are supplied.

a strong significant negative effect ranging between -6.5%-points with product fixed effects and -5.9%-points with market-by-date fixed effects. Since it is the prescribing physician that browses through products in alphabetical order on the computer screen, there would be no reason to expect an effect on consumer choice or pricing, were the physician's choice not affected by the product rank.

As shown in Table 5, the marginal effect of rank on prescriptions is negative for all levels of competition, starting at -5.2%-points with 2 firms to -1.2%-points with 8 firms.

Table 6 Panel B shows results for the effect of alphabetical rank on market share. This investigates to what extent a prescription gets converted into purchase in the Danish system. Again we find a significant and negative main effect ranging between -2.3%-points with market-by-date fixed effects and 3.9%-points with only company and period fixed effects. These results document that the physician's choice of product from the drop-down list passes through to the consumer's purchase decision. A patient is more likely to buy a product simply because it is on the prescription. In Table 5, we

TABLE 6. Regression Results for Four Outcome Variables

	(1)	(2)	(3)	(4)	(5)
Panel A — Outcome: Prescription share					
Alphabetical Rank	-0.0340***	-0.0551***	-0.0582***	-0.0648***	-0.0594***
	(0.00968)	(0.00714)	(0.00782)	(0.00673)	(0.00825)
Alphabetical Rank×No. Firms	0.00155	0.00543***	0.00588***	0.00661***	0.00622***
	(0.00146)	(0.00104)	(0.00111)	(0.000794)	(0.00118)
Panel B — Outcome: Market share					
Alphabetical Rank	-0.0227***	-0.0393***	-0.0257***	-0.0263***	-0.0230***
	(0.00634)	(0.00577)	(0.00630)	(0.00560)	(0.00669)
Alphabetical Rank×No. Firms	0.00237*	0.00404***	0.00261**	0.00146*	0.00223*
	(0.000968)	(0.000858)	(0.000916)	(0.000721)	(0.000974)
Panel C — Outcome: Log price					
Alphabetical Rank	-0.628***	-0.376***	-0.0433**	-0.0696***	-0.0455**
	(0.0630)	(0.0508)	(0.0158)	(0.0168)	(0.0166)
Alphabetical Rank×No. Firms	0.0760***	0.0440***	0.00526	0.00845***	0.00652*
	(0.00979)	(0.00800)	(0.00286)	(0.00199)	(0.00303)
Panel D — Outcome: Revenue share					
Alphabetical Rank	-0.0222***	-0.0346***	-0.0295***	-0.0216***	-0.0288***
	(0.00625)	(0.00582)	(0.00650)	(0.00544)	(0.00693)
Alphabetical Rank×No. Firms	0.00228*	0.00368***	0.00319**	0.00190*	0.00306**
	(0.00101)	(0.000909)	(0.000989)	(0.000776)	(0.00106)
Drug age FE	Yes	Yes	Yes	Yes	No
No. Firms FE	Yes	Yes	Yes	Yes	No
Company FE	No	Yes	Yes	No	Yes
Period FE	No	Yes	Yes	Yes	No
Market FE	No	No	Yes	No	No
Product FE	No	No	No	Yes	No
Market-by-date FE	No	No	No	No	Yes
Clusters	1573	1573	1567	1563	1492
Observations	749859	749857	749851	749741	735396

Note: Standard errors clustered at the market level, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$. Sample includes both branded and generic products. "Company FE" are common for all products in the same pharmaceutical company's portfolio.

see that the marginal effect of rank on market share is always negative, ranging from -.023 with 2 firms to -0.015 with 8 firms.

Note that the effects estimated above include any competitive response to prominence. As we shall see below, prices do indeed respond, so what we have measured is the full equilibrium effect, which is also the most policy relevant.

Table 6 Panel C shows how prices are affected by prominence. Qualitatively, all specifications robustly show the same result: prices are generally downward sloping in rank, but the slope flattens with more firms. In the specification with product-level fixed effects, the price-rank gradient is zero when the maximum number of firms (8) are present. This is consistent with what we saw in the raw averages in Figure A.1d. The quantitative magnitudes of the pooled OLS results are very large, consistent with the raw averages, but as soon as we add company and market fixed effects, the magnitudes fall to much more plausible levels. The main effect ranges from 4.3 to 7.0%-points per rank, diminishing by 0.5 to 0.8%-points per extra firm active. With product-level fixed effects, the effect goes from 5.3%-points in duopoly markets to 0.2%-points with 8 firms (Table 5).

Table 6 Panel D shows the effect of rank on revenue share.[18] Since market shares are decreasing in alphabetical rank and prices are either decreasing or mostly flat, it is perhaps not surprising that the firm's revenue as a fraction of total market revenue is also decreasing in alphabetical rank, with a pattern similar to that of the market share. Revenue share decreases between 2.2%-point in alphabetical position in the specification with product-level fixed effects and 4.4%-point in the specification with company fixed effects only. Hence, in equilibrium it is a profitable advantage for firms to be ranked early in the alphabet.

18. We use the revenue share as opposed to log revenue due to the prevalence of zero sales in our data.

4.3. Robustness

We subject the reduced-form results to three checks. First, we ask whether the mechanism is equally strong in markets where the branded is before versus after the generics in the alphabetical ordering. Second, we restrict the sample to generic products only. Third, we exclude products that enter or exit during the sample period.

Branded first vs. last. Tables A.9 and A.10 split the price-on-rank regression according to whether the first-ranked product is a generic or not. This relates to the institutional detail that the originator's branded name is either before or after all the generics in the system. If the estimated effect reflects prominence rather than some artefact of product composition, it should be strongest where a generic occupies the prominent first position, because that is where rank can steer prescriptions between close substitutes. Table A.9 summarizes the effect on the key variable, price, and shows it for the full sample, the branded first, and generic first subsamples. The effect is -13.3% and statistically significant when the generic is first, but is insignificant otherwise. Table A.10 shows the same four panels as in Table 6, but for the subsample where the generic is first, and shows that the estimate varies from insignificant to -13.3% across the key specifications. We conclude that there is some evidence for our key mechanism, even though the estimate varies somewhat in the subsample where the generics are first.

Generic products only. Table A.8 reports the main specifications estimated on generic products alone. This checks if generic products respond differently to alphabetical rank than branded products. Thus, we impose two extra restrictions (see Table A.2): first, to drop all non-generic products, and second to drop market-periods with less than two generic products competing. The latter ensures that there is active competition among generics, and it is the basis for our counterfactual results later. Our sample is thus reduced to the 293,075 product-periods where the product's name contains quotation marks, so that we are absolutely certain it is a generic. The effect on prescription share ranges from -11.3% to -17.8% per rank (statistically significant), compared to -3.4% to -6.5% for the full sample. The effect on price once again is dramatically larger in columns (1) and (2) with fewer FE, but drops to between -3.7% and -4.9% per rank in Table A.8, compared to between -4.3% and -7.0% for (3)-(5) in the primary results in Table 6. So the effect on prescriptions is stronger, and the effect on price is similar.

Excluding products that enter or exit during the sample. Table A.11 reports the main specifications estimated after dropping every product that enters or exits within the sample window. Because identification comes from changes in incumbents' ranks, and those changes are caused by the entry and exit of competitors sorting earlier in the alphabet, one might worry that the estimates are driven by the churning products themselves - for instance if entrants systematically target prominent positions. Dropping those products leaves only incumbents present throughout, whose rank variation is inherited entirely from others' churn. Details are in Appendix A.3. Overall, we find very robust results for the effect on prescription share. For price, we lose a lot of statistical precision, especially with the more fine-grained product or market-by-date FE, but the point estimates are stable and lie within the confidence intervals of the preferred specification. We conclude that the results are not driven by something particular to the products that enter or exit themselves, although those products naturally contribute a large share of the variation in rank.

5. Demand

This section specifies the demand model: the physician's prescription and the consumer's purchase. We outline the shared search framework, specify each agent's choice problem, then present the estimates.

5.1. A Structural Model of Ordered Search

The structural search model we use is essentially a minor adaptation of Moraga-González et al. (2023) to a setting with single-good search as opposed to bulk search in their case.[19] An agent i faces a discrete set of products, $\mathscr{J}$, where product j yields consumption utility

$$u_{ij} = \delta_{ij} + \varepsilon_{ij}, \quad \varepsilon_{ij} \sim \text{IID} F_0^{\varepsilon},$$

where the CDF F_0^{ε} is taken to be Type 1 Extreme value, meaning that u_{ij} has CDF $F_{ij}^{u}(z) = F_0^{\varepsilon}(z - \delta_{ij}) = \exp[-\exp(\delta_{ij} - z)]$. We assume that δ_{ij} is known prior to search but that in order to observe ε_{ij}, the agent must incur a search cost $c_{ij} \sim \text{IID} F_{ij}^{c}(\cdot)$. Search costs are indexed by a shifter, μ_{ij}, $F_{ij}^{c}(c) = F_0^{c}(c|\mu_{ij})$. There is free recall, meaning that all previously searched products are remembered.

This dynamic problem can be solved using the index-based "Pandora's rule" of Weitzman (1979). Consider a situation where maximum utility the agent has encountered thus far is r. The expected marginal gain from searching product j is

19. In their setting, a consumer must choose which dealer to visit. Upon visiting a dealer, all products there are immediately searched together. If all dealers had only one product, their model would simplify to ours.

then

$$H_{ij}(r) = \mathbb{E}[\max(u_{ij} - r, 0)] = H_0(r - \delta_{ij}), \quad \text{where } H_0(r) \equiv \int_r^\infty (z - r)\,\mathrm{d}F_0^\varepsilon(z).$$

Given that ε_{ij} has full support, $H_{ij}(\cdot)$ is a strictly decreasing function with a well-defined inverse. We therefore define the *reservation utility*, r_{ij}, as the value that makes the agent indifferent between searching j or not given search costs, c_{ij}:

$$r_{ij} = H_{ij}^{-1}(c_{ij}) = \delta_{ij} + H_0^{-1}(c_{ij}).$$

Weitzman (1979) showed that the optimal strategy is to inspect products in descending order of reservation utilities, until an option is encountered with realized utility, u_{ij}, above all remaining reservation utilities. To avoid enumerating the many possible search paths, we may use a result by Armstrong (2017) and Choi et al. (2018), who showed that the problem can be recast as a static discrete choice problem with utilities

$$w_{ij} = \min\{r_{ij}, u_{ij}\},$$

so that the final choice is the product with the highest index w_{ij}. The final purchase probabilities follow from the CDF of w_{ij} from the model primitives.

The reservation utilities have CDF

$$F_{ij}^r(r) = \Pr\left[H_{ij}^{-1}(c_{ij}) \le r\right] = 1 - F_{ij}^c\left[H_0(r - \delta_{ij})\right].$$

Thus, the CDF of w_{ij} is

$$F_{ij}^w(w) \quad = \quad \Pr\left[\min\{r_{ij}, u_{ij}\} \le w\right] = 1 - F_{ij}^c\left[H_0(w - \delta_{ij})\right]\left[1 - F_{ij}^u(w)\right].$$

Moraga-González et al. (2023) then prove that if we assume

$$F_{ij}^{c}(c) = \frac{1 - \exp\left\{-\exp\left[-H_0^{-1}(c) - \mu_{ij}\right]\right\}}{1 - \exp\left\{-\exp\left[-H_0^{-1}(c)\right]\right\}},$$

then w_{ij} becomes T1EV with location parameter $\delta_{ij} - \mu_{ij}$,[20] so that the final choice probabilities take the logit form

$$\Pr(i \text{ buys } j) = \frac{\exp(\delta_{ij} - \mu_{ij})}{\sum_{k \in \mathscr{J}} \exp(\delta_{ik} - \mu_{ik})}.$$

With regard to econometric identification, the logit choice probabilities mean that standard intuition from discrete choice applies. However, because δ_{ij} and μ_{ij} enter choice probabilities only through the difference $\delta_{ij} - \mu_{ij}$, they are not separately identified from choice data alone. A covariate placed in search costs and the same covariate placed with the opposite sign in consumption utility yield identical likelihoods, identical predicted shares, and hence identical firm-perceived demand. Nevertheless, it may change search decisions and hence consumer welfare. But absent direct data on search, the allocation of a covariate is therefore a normalization rather than a testable restriction (Moraga-González et al., 2023). Two implications deserve emphasis.

First, our specification does not assume that branded and generic products with the same molecule are perceived as homogeneous. The generic dummy in μ_{ij}^{c} is observationally equivalent to a vertical differential of equal magnitude in δ_{ij}^{c}; and both are consistent with the branded preference documented in e.g. Arcidiacono

20. That is, $F_{ij}^{w}(z) = \exp\left\{-\exp\left[-(z - \delta_{ij} + \mu_{ij})\right]\right\}$.

et al. (2013) and Dubois and Lasio (2018). Whether the preference reflects taste or information is not identified from our data, and we take no stand on it, because our counterfactuals do not depend on that and we will not compute consumer welfare, but only expenditures.

Second, two covariates do require a substantive stand: the alphabetical rank R_{ij}, which our counterfactuals reorder, and the dummy for the pharmacist's recommendation to the consumer, which is a discontinuous function of the price vector. We assume that both are purely informational: the ranking is not observed by the consumer and thus cannot enter her consumption utility, and it is just a function of alphabetical ordering of company names. The pharmacist's recommendation is not a consumption attribute of the product, and the utility of outside consumption should not be discontinuous in the price vector.

Note also that there is no outside good in the model, since we do not observe prescriptions that were not filled. However, consumers are always reimbursed by the government based on the cheapest product in the market, so out-of-pocket costs do not rise nearly as much. Conversely, the price difference between the cheapest product and alternative products is paid fully out of pocket to incentivize substitution to the cheapest product.

5.2. The Consumer's Purchase Choice

The consumer enters the pharmacy with a prescription and looks for a product that maximizes her indirect utility

$$\delta^c_{ij} = \beta^c_{1i} \log p_{ij}, \qquad \beta^c_{1i} = \beta^c_1 + \gamma^c x_i,$$

where p_{ij} is the consumer-specific pharmacy retail price of product j, and x_i denotes consumer characteristics (income), so that the marginal utility of money is allowed to vary with income. The consumer is subject to search cost, affected both by the product on the prescription and what the pharmacy recommends. The search cost shifter is modeled as

$$\begin{aligned}\mu_{ij}^c &= \beta_2^c \mathbf{1}_{\{p_{ij}\in\mathscr{A}(p)\}} + \beta_3^c \mathbf{1}_{\{p_{ij}\in\mathscr{B}(p)\}} + \beta_{4i}^c \mathbf{1}_{\{a_i=j\}} \\ &\quad + \beta_5^c \mathbf{1}_{\{a_i=j\wedge p_{ij}\in\mathscr{A}(p)\}} + \beta_6^c \mathbf{1}_{\{a_i=j\wedge p_{ij}\in\mathscr{B}(p)\}} + \beta_7^c \mathbf{z}_{ij}^c,\end{aligned}$$

where p denotes the full vector of prices, $p = (p_{i1}, ..., p_{iJ})$, $a_i \in \mathscr{J}$ denotes the prescribed product for consumer i, and $\mathbf{z}_{ij}^c$ collects additional search cost shifters: dummies for generic, parallel import, and delivery failure. The coefficient on the prescription dummy, $\beta_{4i}^c = \beta_4^c + \sigma^c \nu_i$ with $\nu_i \overset{\text{iid}}{\sim} N(0,1)$, is allowed to vary across consumers, capturing that different patients place different weight on the information conveyed by the prescription. The “ABC” sets form a partition based on the regulation governing pharmacists outlined in Section 2.1. Specifically, if we let $\underline{p} = \min_j p_{ij}$

$$\mathscr{A}(p) = \{\underline{p}\}, \mathscr{B}(p) = (\underline{p}; 1.05\underline{p}], \mathscr{C}(p) = (1.05\underline{p}; \infty).$$

The rules regarding generic substitution depend on the price of the prescribed product: If the prescribed product a_i has $p_{ia_i} \in \mathscr{B}(p)$, then the law merely *encourages* substitution, while for prescribed products with $p_{ia_i} \in \mathscr{C}(p)$, the pharmacist *must* recommend substitution. The recommended product should always be one with $p_{ij} \in \mathscr{A}(p)$.

We identify consumption utility parameters, δ_{ij}^{c}, separately from the role of the search cost, μ_{ij}^{c}, using the assumption that consumer utility is continuous in expenditure, whereas the pharmacist's recommendation is discontinuous in price. Thereby, any discontinuities in final purchase probabilities at the cutoffs of the ABC-regions must be due to the pharmacist affecting search costs. The most important one is β_2^c: a large negative value indicates that the consumer is almost always aware of the cheapest product. Since Figure 3 showed a dramatic discontinuity at the minimum price (the market share was nearly cut in half at even tiny price increments above the minimum), we may expect a large coefficient.

The coefficients $(\beta_4^c, \beta_5^c, \beta_6^c)$ capture the effect of the prescription itself. Some consumers may simply trust the physician's choice over the pharmacist's recommendation, and given how strongly the alphabetical ordering shifts physician choices, such trust would be unwarranted. Regardless, any effect of the alphabetical rank on final demand must work through this channel because the consumer does not see the ranking. That is, the products ranked first alphabetically are chosen more by physicians and then consumers remain loyal to that prescription, if $(\beta_4^c, \beta_5^c, \beta_6^c)$ are large in magnitude.

5.3. The Physician's Prescription Choice

We model physicians as agents for the patients: they internalize the expected out-of-pocket expense their prescription imposes on the patient, and choose the product that minimizes it.

$$\delta_{ij}^{p} = \omega \mathbb{E}(\log p_{ik} | a_i = j) = \omega \sum_{k \in \mathscr{J}} \Pr_i(k | a_i = j) \log p_{ik},$$

where $\Pr_i(k|a_i = j)$ is the probability that the consumer buys k with the prescription $a_i = j$. Physicians, like consumers, face search costs, which we model as

$$\mu_{ij}^{p} = \beta^{R} \log R_{ij} + \log R_{ij} \times \gamma^{R} \mathbf{x}_i + \beta^{p} \mathbf{z}_{ij},$$

where $R_{ij} \in \{1, ..., J\}$ is the alphabetical rank of product j, entering in logs; $\mathbf{x}_i$ is a vector of patient characteristics (gender, age, log income) interacted with $\log R_{ij}$; and $\mathbf{z}_{ij}$ is a vector of additional search cost shifters, which varies across specifications but includes a dummy for generic (interacted with years since patent expiration) and a dummy for parallel import.

5.4. Estimation and Identification

Let i denote a transaction with prescription a_i and final patient choice j_i. Given the assumptions made thus far, the likelihood for observation i is

$$\begin{aligned} L_i(\theta) &= \Pr(a_i \text{ prescribed}) \Pr(j_i \text{ chosen} | a_i) \\ &= \frac{\exp(\delta_{ia_i}^{p} - \mu_{ia_i}^{p})}{\sum_{k \in \mathscr{J}_i} \exp(\delta_{ik}^{p} - \mu_{ik}^{p})} \int \frac{\exp[\delta_{ij_i}^{c} - \mu_{ij_i}^{c}(\nu)]}{\sum_{k \in \mathscr{J}_i} \exp[\delta_{ik}^{c} - \mu_{ik}^{c}(\nu)]} \varphi(\nu) \, d\nu. \end{aligned}$$

where $\varphi(\cdot)$ is the standard normal density and $\mu_{ik}^{c}(\nu)$ denotes μ_{ik}^{c} evaluated at $\beta_{4i}^{c} = \beta_4^{c} + \sigma^{c} \nu$. Given that we observe a_i for all transactions, we can estimate the model in two steps: first estimating the parameters governing the consumer behavior; then computing expected consumer expenditures and holding those fixed and estimating physician preferences. The rest of this section discusses our key identifying assumptions and what variation in the data pins down each coefficient.

One crucial assumption is that physicians do not condition their prescription choice on the realization of the consumer logit error terms, nor on the consumer's realized draw of the random coefficient v_i governing her trust in the prescription, which limits how useful physicians can actually be to consumers. The assumption is warranted here because generics must be scientifically proven to yield similar physiological results to the original; it would not hold if we instead studied physician choices *across* different molecules.

A related concern is selection: price-sensitive consumers might select physicians who place more weight on expenditures and have lower search costs. This seems unlikely here because the prescription does not restrict the consumer's purchase choice, unlike a setting with *different* molecules, where the prescription is a legal barrier forcing the consumer to buy within that market. Put differently: unwanted effects of the prescription are purely search frictions, undone by a single word to the pharmacist, whereas changing physician requires incurring a fee and going through some bureaucratic hassle, and there is very limited information to consumers on alternative physicians.[21]

Price endogeneity is a primary concern in any demand estimation. Here, the demand discontinuity implies that the pricing equilibrium is a distribution over prices

21. Changing physician costs a small amount, 150 DKK, whereby the patient can choose a physician while only observing the age, name, and address of the clinic (Huang and Ullrich, 2026). Thus, any information about physician prescribing behavior must come through e.g. word-of-mouth.

rather than a price vector (Section 6), which generates substantial within-product price variation over time. Because that variation is inherently unpredictable in equilibrium, it is exogenous to demand shocks.

5.5. Results

Table 7 shows estimates for the consumer model, which includes both observed heterogeneity – through the interaction between income and price – as well as unobserved heterogeneity – in the form of a random coefficient on the prescription dummy. We start by focusing on the coefficients other than these heterogeneity terms.

The results show that consumers respond strongly to out-of-pocket expenditures. For search costs, we see an extremely powerful effect of generic substitution: search costs for the cheapest product (in the $\mathscr{A}$-region) are about 2 utils lower than products that are at least 5% more expensive (the $\mathscr{C}$ region, the reference category); evaluated at the mean log income, this is nearly equivalent to a threefold increase in out-of-pocket expenditures.

If the product is prescribed, it also entails far lower search costs, likely because the pharmacist has to explain to the consumer about this product, which makes it salient compared to products that are neither cheapest nor prescribed. However, we see that the strongest effect of the prescription occurs for the high price region. This means that demand going to products in the most expensive $\mathscr{C}$ region is likely to a large degree only explained by prescriptions. That is, the loyal demand a firm has is largely derived from prescriptions.

Finally, the remaining coefficients reveal consumer heterogeneity in demand. First, we find that richer consumers are *more* price sensitive. While puzzling from a consumption-savings perspective, richer individuals may have different information about pharmaceuticals affecting their trust in generic substitution. Second, we see that there is a considerable amount of heterogeneity in the prescription dummy: the effect is -0.91, but when the prescribed is cheapest, it is even stronger at -1.20. However, the dispersion around that mean is massive: with a standard deviation of 1.73, 90% of consumers fall within a wide range from -3.76 to 1.95. This means that some consumers are extremely attentive to what the physician prescribed, and they will with a very high likelihood be taking that product actively into consideration, whereas other consumers are the opposite.[22]

Table 8 presents the estimates from the physician choice model. We see that the physician internalizes her impact on the consumer's decision, because there is a significant and large coefficient on the expected consumer expense. With regard to search costs, we see a strong increase in search costs down the alphabetical rank, meaning that the physician is most likely to start at the top. However, the interactions reveal that this effect is smaller for females, implying that she is likely to spend more time clicking through products in search of a good match for females than males. Similarly, the effect of rank on search costs is lower for the young and the poor, implying that the physician spends more time for those demographic groups.

The dummy for a generic product is positive and large, implying that the physician is less likely to explore generic options. However, the interaction between the generic dummy and the number of years since patent expiration is negative, implying that as

22. We have experimented with random coefficients on other variables (price and the generic dummy). We found the greatest degree of heterogeneity for the prescription dummy, and since we are interested in the prescription-loyal demand generated by rank, this is the specification we have chosen.

TABLE 7. Estimates: Consumer Purchase Choice

	Estimate	s.e.
$\log p_{ij}$	-1.452***	0.0477
$\log(\text{income}_i) \times \log p_{ij}$	-.0249***	0.0040
Search cost shifters (μ^c_{ij})		
$\mathbf{1}_{p_{ij}\in\mathscr{A}(p)}$	-1.8909***	0.0040
$\mathbf{1}_{p_{ij}\in\mathscr{B}(p)}$	-0.8552***	0.0056
1(prescribed)	-0.9055***	0.0113
1(prescribed) $\times\ \mathbf{1}_{p_{ij}\in\mathscr{A}(p)}$	-0.2906***	0.0199
1(prescribed) $\times \mathbf{1}_{p_{ij}\in\mathscr{B}(p)}$	-0.0180	0.0133
1(generic)	0.0990***	0.0049
1(parallel import)	0.5536***	0.0146
1(delivery failure)	1.3195***	0.0066
Random coefficient: $\sigma(1(\text{prescribed}))$	1.7346***	0.0182
Own-price elasticity, avg.	-1.2389	
Own-price elasticity, std. dev.	0.2209	
N		967,914

Notes: Standard errors are based on the Sandwich Formula. Significance levels: * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$. Coefficients in the lower block enter the search cost shifter μ^c_{ij}, so that a negative estimate indicates lower search costs. The own-price elasticities are computed holding fixed the effects of generic substitution (the discontinuity at the minimum price), and they integrate out the random coefficient, and average over consumers, and then take the unweighted mean over products.

time goes, physicians become more likely to consider generic products compared to branded products. We also note a strong positive coefficient on the dummy for parallel imports, implying that the physician is less likely to explore those products as well.

6. Supply

We model pricing as a game of incomplete information, in which firms receive private payoff shocks and submit a price. Given estimated demand and cost, the resulting equilibrium is characterized by a distribution of prices. This is a direct consequence of the undercutting motive, rather than of cost dispersion, which is what a standard markup inversion would attribute it to (Berry et al., 1995). We therefore recover marginal cost from the location and shape of the price distribution instead.

TABLE 8. Estimates: Physician Prescription Choice

	Estimate	s.e.
Expected consumer expense (log)	-0.2922***	.0363
Search cost shifters (μ^p_{ij})		
1(generic)	1.0880***	.0073
1(generic) $\times$ years since patent expiry	-0.1919***	.0035
1(parallel import)	0.8774***	.0234
1(parallel import) $\times$ 1(generic)	0.0819**	.0254
$\log R_{ij}$	0.4297***	.0165
$1(\text{female}_i) \times \log R_{ij}$	-.0580***	.0031
$\text{age}_i \times \log R_{ij}$	0.00092***	.000094
$\log(\text{income}_i) \times \log R_{ij}$	0.0158***	.00013
Own expense elasticity, avg.	-0.2265	
Own-price elasticity, avg.	-0.0368	
N		967,914

Notes: See Table 7; coefficients in the lower block again enter μ^p_{ij}, so that a negative estimate indicates lower search costs. The own expense elasticity is computed numerically by adding $\log(1.01)$ to the expected consumer expense conditional on prescribing j. The own-price elasticity is computed by first increasing the price of j by 1%, then recomputing expected consumer expenses, and then computing prescription probabilities. Since the elasticity of expenditures wrt. price is considerably below 1, the own-price elasticity is smaller in magnitude than the own expense elasticity.

Appendix C shows why the standard markup inversion fails at the magnitude of demand discontinuity we estimate. Applied to a simplified version of our demand system at the estimated parameters, some firm can profitably undercut the price leader in 97.6% of the market-periods in our sample.

6.1. The Pricing Game

We consider that the market consist of two types of firms: generic firms and branded firms. We model generic firms as strategic players who simultaneously choose prices every two-week period to maximize static profits. Firm j earns

$$\pi_j(\mathbf{p}) = (p_j - c_j)\,s_j(\mathbf{p}),$$

where $c_j \geq 0$ is marginal cost and $s_j(\mathbf{p})$ is the market share implied by the estimated demand system, integrating out physician prescriptions and consumer purchases:

$$s_j(\mathbf{p}) = \sum_{i \in \mathscr{I}} \sum_{k \in \mathscr{J}} \Pr(a_i = k; \mathbf{p}) \Pr(j_i = j \mid a_i = k; \mathbf{p}).$$

We assume that branded firms are non-strategic, using the pricing distributions we observe in the data. This assumption is motivated by empirical findings that branded prices respond far less to incentives than generic prices do. See e.g. Vandoros and Kanavos (2013) who find that branded prices barely fall in response to generic entry, and Kaiser et al. (2014) who study the response to a reform of the reimbursement system in Denmark, where branded prices fell by only 2% against 36% for generics and 28% for parallel imports.[23]

We define a *market configuration* as a tuple describing the number and types of firms ranked by alphabetical position, e.g. $(\mathrm{G}, \mathrm{G}, \mathrm{B})$ denotes a three-firm market where the two alphabetically first products are generic and the third is branded. We solve the pricing game separately for each configuration, allowing brand presence and position to shape the equilibrium — for instance, the competitive dynamics differ when the branded product is listed first, capturing more prescriptions, versus last.

23. We speculate that international reference pricing may be a reason for the weak response: Denmark appears in the basket of 15 of 31 European countries, and such mechanisms tend to emphasize branded prices even if generics are present (Toumi et al., 2014).

6.2. Equilibrium

We model the pricing game as one of simultaneous moves with incomplete information. Each generic firm j chooses a price from a discrete grid $\mathscr{P}_j$ to maximize expected profits plus a private payoff shock, given beliefs about competitor strategies, σ_{-j}:

$$p_j^* = \arg\max_{p \in \mathscr{P}_j} \mathbb{E}_{p_{-j}}[\pi_j(p, p_{-j})] + \lambda\, \varepsilon_j(p), \qquad \varepsilon_j(p) \sim \text{IID T1EV}. \tag{2}$$

The expectation is over rival strategies and $\lambda > 0$ governs the importance of private information relative to observable strategic payoffs. The IID Type 1 Extreme Value shocks capture idiosyncratic factors that shift the relative attractiveness of different price points across pricing periods — cost fluctuations, inventory considerations, dynamic incentives the static model abstracts from, and other firm-specific motives.

The T1EV assumption gives firm j's *best-response distribution* the logit form:

$$\sigma_j(p \mid \sigma_{-j}; \lambda) = \frac{\exp\{\mathbb{E}_{p_{-j}}[\pi_j(p, p_{-j})]/\lambda\}}{\sum_{q \in \mathscr{P}_j} \exp\{\mathbb{E}_{p_{-j}}[\pi_j(q, p_{-j})]/\lambda\}}. \tag{3}$$

We read σ_j as a best response rather than as a randomization device: conditional on its realized shocks, firm j plays a pure strategy; σ_j is the distribution of that pure strategy induced by the shocks, and it is what rivals can condition on. A Bayesian Nash Equilibrium (BNE) is a strategy profile $\sigma^* = (\sigma_1^*, \ldots, \sigma_J^*)$ such that (3) holds simultaneously for all strategic firms, while non-strategic firms play according to their empirical pricing distributions.[24]

λ governs the degree of dispersion in equilibrium play, interpolating between two limits. As $\lambda \to 0$, the BNE converges to an equilibrium of the complete-information game, which we argue below must be in mixed strategies.[25] As $\lambda \to \infty$, the BNE converges to uniform randomization by all players. With λ finite, the BNE is somewhere in between. The shape comes from the logit functional form (3), marginal cost, and the shape of demand.

To establish some intuition for why the equilibrium is dispersed, consider first the limit $\lambda \to 0$, in which private information is negligible and (3) becomes the complete-information best response. If competitors set a fixed price, then because of the discontinuity in demand, j's best response is either to undercut to capture the generic substitution bonus, or to hike the price to exploit prescription-loyal demand. Matching is always dominated by undercutting (Proposition C.2), and asymmetric equilibria invite an undercut in virtually all observed markets, given our estimates (Proposition C.1).

With $\lambda > 0$ the same argument rules out degenerate beliefs. Suppose firm j believed its rivals would price at a known $\bar{p}$. Then $\mathbb{E}_{p_{-j}}[\pi_j(p, p_{-j})]$ again jumps below

24. Our equilibrium concept is equivalent to the *Quantal Response Equilibrium* (QRE) of McKelvey and Palfrey (1995). We prefer the BNE interpretation because there are several reasons to think that firms have pricing motives beyond pure profit given our simplified model.

25. For the continuous game. When the game is discretized, a pure strategy equilibrium arises with full mass on first grid point with positive profits. That equilibrium is an artificial consequence of the fact that discretization eliminates a profitable deviation there. And our estimated λ is far away from that.

$\bar{p}$ and j's (logit) best response concentrates immediately below $\bar{p}$ — at which point $\bar{p}$ is dominated for the rivals, who would themselves undercut. Degenerate beliefs are therefore never mutually consistent, so equilibrium dispersion survives at any λ.

Regardless of λ, competitor prices must be unpredictable in equilibrium — what differs is the source and interpretation of that unpredictability. By (3), how spread out σ_j is depends on the variation of $\mathbb{E}_{p_{-j}}[\pi_j(p, p_{-j})]$ across $p \in \mathscr{P}_j$ relative to λ: obviously, if $\lambda \to \infty$, then σ_j tends to the uniform. Conversely, in the extreme where expected profit is exactly flat in p, even a vanishingly small λ also yields a uniform best response. So dispersion in the best response can arise either from a large λ or from the shape of $\mathbb{E}_{p_{-j}}[\pi_j(p, p_{-j})]$, which depends on σ_{-j}. And note that expected profit cannot be flat unless σ_{-j} is itself dispersed: otherwise competitor prices are predictable, and j's best response concentrates on undercutting (or hiking). Competitor prices are unpredictable in both limits, but for different reasons: idiosyncratic shocks, or mutual near-indifference.[26] What our estimates of λ measure is where between these two the data place each market configuration. The familiar case in which each firm's price is a deterministic function of marginal cost requires $\lambda = 0$ *together with* a differentiable demand system; our framework nests it and lets the data reject it.

Equilibrium computation. To solve for the BNE for a given market configuration and pair (c, λ), we use the homotopy method of Turocy (2005) as implemented in `Gambit` (McKelvey et al., 2006).[27] The homotopy traces the path of solutions starting

26. These two interpretations are exactly at the heart of the Purification Theorem of Harsanyi (1973).

27. We use the solver `logit_atlambda` from `pygambit.nash`.

from uniform randomization ("$\lambda = \infty$") and following the path of equilibria as λ is decreased to the chosen value.

6.3. Estimation

The supply-side primitives are marginal cost, c, and the private information scale, λ. We allow both parameters to vary freely across market configurations, but assume that they are identical for all generic firms in a given configuration.[28]

6.3.1. The Estimator. To estimate the model, we search for the values of (c, λ) so that the equilibrium price distributions match the observed price distribution for the given market configuration as closely as possible. We compute the sum of squared differences on the pricing grid for each strategic firm and then sum over firms. Let $\sigma_j^{\text{data}}(p)$ denote the observed frequency of prices at grid point $p \in \mathscr{P}_j$ within a market configuration, and let $\sigma_j(p \mid c, \lambda)$ be the BNE. We then estimate

$$(\hat{c}, \hat{\lambda}) = \arg \min_{c \geq 0, \lambda > 0} \sum_{j=1}^{J} \sum_{p \in \mathscr{P}_j} \left[\sigma_j^{\text{data}}(p) - \sigma_j(p \mid c, \lambda) \right]^2,$$

where the sum runs over all strategic firms j and grid points p within a configuration. So for each trial value of (c, λ), we must solve the BNE.

6.3.2. Parameter Identification. We start by discussing which features of the observed pricing distribution help pin down the two free parameters.

28. This restriction is motivated by tractability and by the fact that generic firms within a configuration produce the same molecule. Configuration-specific marginal costs allow for the possibility that different competitive environments attract firms with different cost structures.

Marginal cost c primarily determines the *lower bound* of the price support. If $\lambda = 0$, then the support of the equilibrium literally cannot extend below c. When $\lambda > 0$, there is strictly positive support everywhere, but the same intuition applies. Of course, as c increases towards the lowest observed price, the predicted pricing distribution changes shape above that. But for the upper end of the support, marginal cost plays virtually no role. So c is pinned down by the lower tail of the observed distribution.

The private information parameter λ governs the shape of the distribution between two extremes: as $\lambda \to \infty$, the distribution approaches the uniform over the grid. As $\lambda \to 0$, the distribution approaches an equilibrium of the complete information game. For finite λ, the BNE inherits its shape from a combination of the logit functional form (3) and the shape of demand. So λ is pinned down by the estimated demand function and the resulting profit function as well as the overall shape of the observed price distribution.

Beyond c and λ, there are no additional free parameters governing where the pricing distribution is concentrated. The demand curve — specifically, the price elasticity and the strength of generic substitution — determines how compressed the distribution is and where mass accumulates. When a branded product is present, its price (held fixed at the empirical distribution) disciplines generic prices: generics compete below the branded price, and the demand system determines how far below. As a result, the model's predictions about the pricing distribution are tightly disciplined by the estimated demand parameters, and not just (c, λ) in isolation.

One further property of the model is worth noting. When $c = 0$ and log price enters utility, the profit function is homogeneous of degree one: $\pi_j(\kappa \mathbf{p}) = \kappa \pi_j(\mathbf{p})$ for any

$\kappa > 0$.[29] This creates a scale indeterminacy in the continuous game: the equilibrium mixing distribution has a uniquely determined shape, but its location can be scaled freely. In our implementation, the discrete price grid and $\lambda > 0$ jointly pin down the location, which is then identified from the empirical price distribution.

6.3.3. Implementation.

Price grid. We discretize prices using $|\mathscr{P}_j| = 20$ grid points per firm. A fine grid is important: with too few points, exact undercutting is mechanically impossible and the discretized game acquires artificial pure strategy equilibria. Since payoffs are computed on the tensor product grid $\times_j \mathscr{P}_j$, computational complexity is exponential in the number of strategic firms, limiting us to at most four.

Price normalization. To pool market-period observations within a configuration, we normalize each market's prices by the average pre-patent branded price, $\bar{p}_m$. This is without loss of generality for a single market-period and provides a directly interpretable scale: $p = 0.3$ means the product is priced at 30% of the pre-patent price. The grid is equidistant from $p = 0$ to $p = 1$, with one additional point above 1 set to the average observed value of normalized prices exceeding one. To be clear, this means that when a firm j in a given configuration, e.g. the first generic in (G,G,B), sets a price of 10%, we then take all transactions in such configurations and compute the corresponding actual wholesale price, and use the pharmacy fee for that date to calculate the implied retail price. When we compute the fraction of price observations at each grid value, they are weighted the same way, by the number of transactions.

29. This is mainly because our model has no outside option.

TABLE 9. Supply Side Estimates

Market structure	Transactions	Markets	Products	c	λ	$\bar{\pi}$	π^{max}	L
(G, G)	11,909	107	229	0.00	7.75	85.58	104.82	0.18
(B, G, G)	21,017	131	390	0.00	12.80	25.57	35.13	0.27
(G, G, B)	42,081	111	332	0.00	11.65	26.09	34.94	0.25
(G, G, G)	8,827	104	337	0.00	4.99	49.90	62.42	0.20
(G, G, G, G)	16,663	82	393	0.00	1.18	11.87	16.94	0.30
(B, G, G, G)	15,912	117	466	0.00	3.08	15.13	20.97	0.28
(G, G, G, B)	23,962	104	416	0.00	6.35	23.83	35.23	0.32

Note: G denotes Generic, B denotes Branded (non-strategic). A configuration such as "G,G,B" refers to all market-period pairs with three firms where the two alphabetically first products are generic and the third is branded. $\bar{\pi}$ is expected equilibrium profits (absent idiosyncratic shocks), $\pi^{\max}$ is the maximum expected profit a firm could earn when opponents play equilibrium strategies, and L is the share of the maximum profit that is lost from following the equilibrium distribution rather than pure profit maximization. Note that all three measures ignore the private shocks themselves.

6.4. Estimates

Table 9 reports the supply-side estimates.

Marginal cost is estimated at zero for all configurations, on the boundary of the parameter space. This is consistent with the view that the marginal cost of manufacturing a generic pharmaceutical (materials, packaging) is small in absolute terms, while the economically substantial costs — regulatory approval, developing the molecule's formulation — are fixed and, for products already past loss of exclusivity, sunk by the time firms set prices.

The private information parameter λ inherits its units from profits, which are measured in DKK per prescription. To interpret its magnitude, we compute three quantities for each market configuration. Let $\bar{v}_j(p) \equiv \mathbb{E}_{p_{-j}}[\pi_j(p, p_{-j})]$ denote firm j's expected profit from price p, integrating over rivals' equilibrium strategies. Then

$$\pi^{\max} \equiv \frac{1}{J}\sum_j \max_{p\in\mathscr{P}_j} \bar{v}_j(p), \qquad \bar{\pi} \equiv \frac{1}{J}\sum_j \sum_{p\in\mathscr{P}_j} \sigma_j(p \mid \sigma_{-j};\lambda)\,\bar{v}_j(p), \qquad L \equiv 1-\bar{\pi}/\pi^{\max}.$$

Here $\pi^{\max}$ is the largest profit a firm could obtain by always playing the same price against rivals' equilibrium play, and $\bar{\pi}$ is the profit it earns on average in equilibrium, when its price responds to private payoff shocks as well. The difference, L, measures the profit given up because firms' prices are driven by more than the profit component of the objective.[30]

Across market configurations, L ranges from 18% in (G,G) to 32% in (G,G,G,B), with most configurations near 25%. Private considerations are therefore economically material without dominating: firms' prices track expected profit closely enough that the equilibrium distribution remains far from uniform, but loosely enough to sustain the dispersion the data display.

6.5. Model Fit

Figures 4 and 5 compare the model-implied and observed price distributions for each market configuration. In each figure, the left panel shows the empirical price frequencies and the right panel shows the BNE predictions.

It is important to keep in mind what drives the model predictions. With marginal cost estimated at zero, the entire shape of the predicted pricing distribution is determined by the demand system and a single free parameter, λ. The demand curve — including the price elasticity, the generic substitution bonus, and when applicable the branded product's fixed pricing — pins down where prices concentrate and how compressed the distribution is. The parameter λ can only warp between two extremes: uniform randomization over the grid (at $\lambda \to \infty$) and full concentration at the lowest

30. L is not a welfare loss: firms also receive the private payoff $\lambda \varepsilon_j(p)$.

grid point (at $\lambda \to 0$). Any structure in the model's predicted distributions between these extremes — the location of the mode, the right-tail behavior, and the differences across configurations — is a consequence of the estimated demand parameters, not of supply-side degrees of freedom.

With this in mind, several patterns emerge. First, generic prices fall with increasing competition. In pure generic markets, probability mass piles up near the bottom of the support, reflecting intense undercutting competition. Second, the competitive pressure from a branded product is softer than the pressure from a generic product. This is visible when comparing (G,G,G), where prices concentrate below 0.4, to (G,G,B) and (B,G,G), where generic prices spread up toward 0.6–0.8.

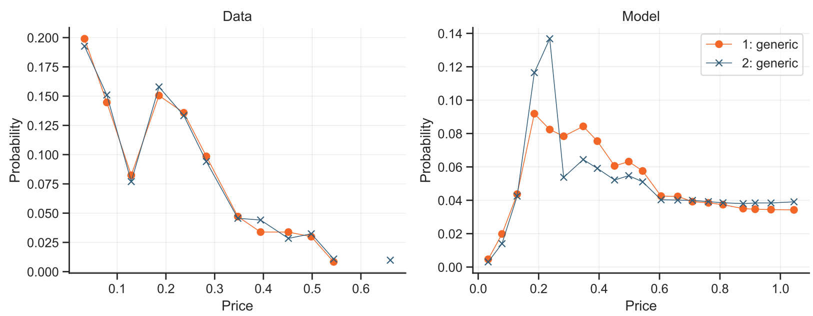
Third, the *position* of the branded product also has an effect. In (B,G,G) markets, the branded product is listed first and captures a disproportionate share of prescriptions through the prominence mechanism. This steers the relatively inelastic prescription-loyal demand toward the branded product, leaving generics to compete for more price-sensitive consumers. As a result, generic prices are slightly lower in (B,G,G) than in (G,G,B), which can be seen in Table 12.

7. Counterfactuals

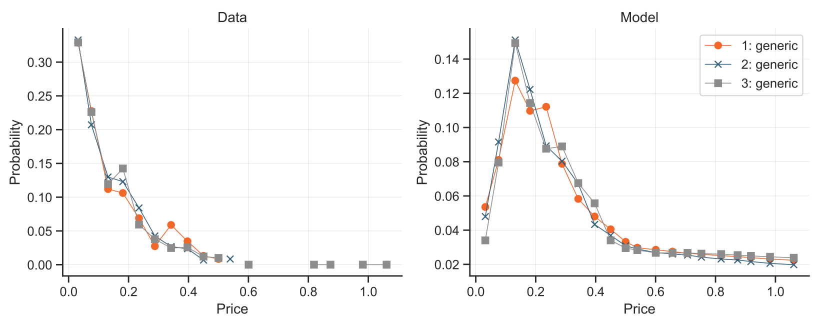
We use the estimated model to three evaluate alternative designs of the physician IT system, alongside the baseline of alphabetical ranking:

1. **No rank effect:** we set $\beta_r^R = 0$ for all r, removing the effect of list position on the physician's choice while leaving all other parameters unchanged. This represents

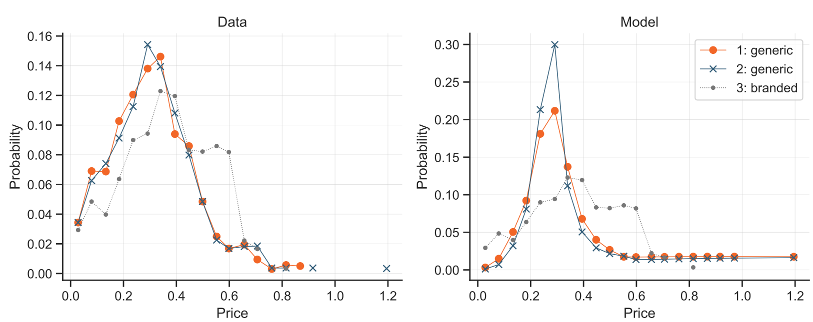
FIGURE 4. Supply Model Fit, $J = 2, 3$



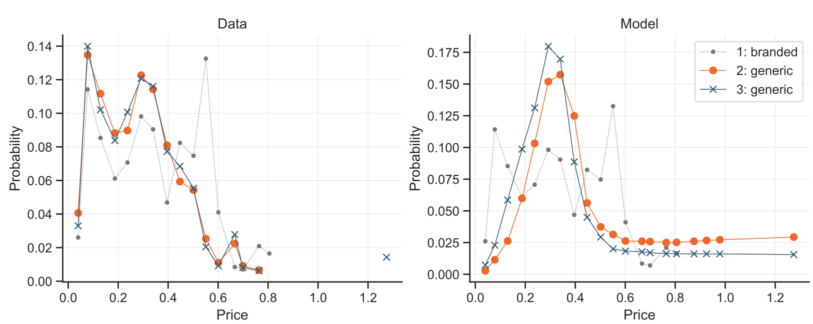
(A) (G, G)



(B) (G, G, G)



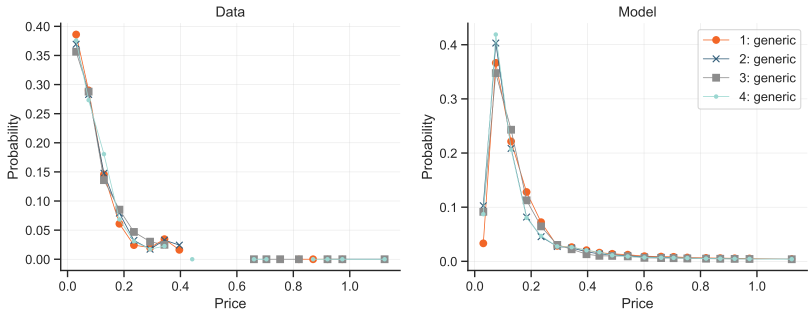
(C) (G, G, B)



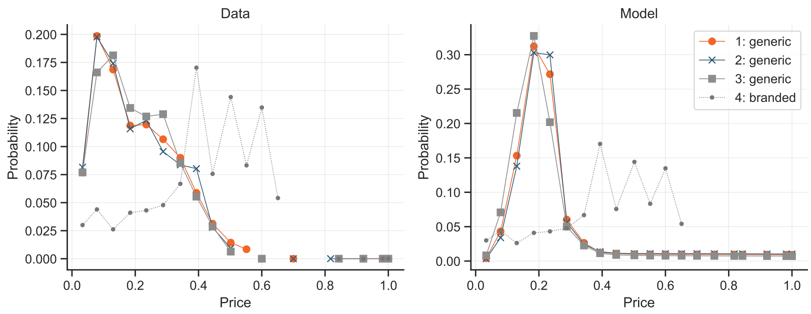
(D) (B, G, G)

Note: Each dot corresponds to a point on the price grid; the y-value is the probability assigned to that price. Prices are normalized by the pre-patent branded price ($p = 1$). Data and model are shown in separate panels because price distributions are very similar across ranks within each panel; the separate format facilitates comparison of the overall distributional shape. The branded product's distribution is the same in both panels because it is treated as non-strategic.

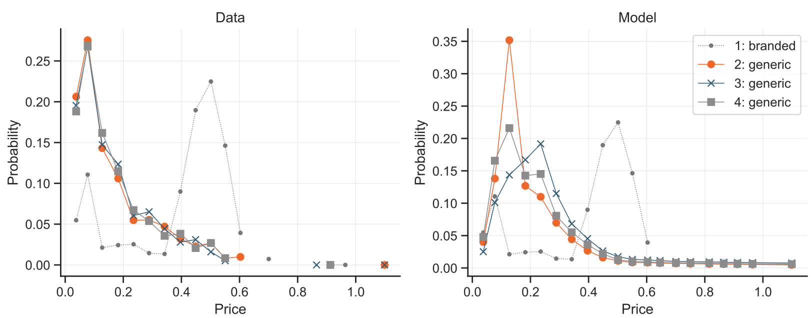
FIGURE 5. Supply Model Fit, $J = 4$



(A) (G, G, G, G)



(B) (G, G, G, B)



(C) (B, G, G, G)

Note: See Figure 4.

a system in which the IT list does not prime the physician toward any particular product.

2. **Price ranking:** we sort the list in ascending order of price, so that the cheapest product is listed first.[31] This entails changing only the sort order in the physician IT system.
3. **Prescribe cheapest:** the physician always prescribes the cheapest product. This is an upper bound on how strongly consumers can be steered through the prescription. While so-called "generic prescribing" has been implemented as an experiment in practice (Patel et al., 2014), it is viewed as a considerable change to the autonomy of physicians, and thus a more dramatic intervention than price ranking.

Each scenario is computed under two pricing regimes. Under *fixed prices*, we hold firms' price distributions at their baseline (model-implied) values and change only the demand system, which recovers the savings from redirecting prescriptions alone. Under *equilibrium prices*, we also re-solve the price equilibrium. Branded firms' prices are held at the observed distributions throughout. Supply parameters (c, λ) are held fixed at the configuration-specific estimates from Table 9.

Three claims summarize the results. First, removing prominence from the IT system leaves aggregate expenditures essentially unchanged (-0.2% in equilibrium). Second, the zero mean effect hides underlying heterogeneity, where the sign of the effect is determined by which firm initially holds prominence. Third, allocating the position by price lowers expenditures by 7.7% in equilibrium against 2.1% at fixed prices, so an evaluation that ignored the pricing response would understate the reform by a factor of 3.7.

31. When two or more products have the same price, we assign the average rank to all tied products.

In all three counterfactuals, alphabetical rank no longer plays a role, which is why expected prices equalize over positions in Table 12. Note, however, that the expected demand faced by a given firm averages over all products in our data matching the market configuration and position, so product-specific characteristics still differentiate firms. This is why prices do not equalize exactly.

7.1. Removing prominence

Eliminating the rank effect has almost no effect on average expenditures. The change is -0.1% at fixed prices (Table 10) and -0.2% in equilibrium (Table 11, column 2). So a system that allocates prominence on an irrelevant factor, the firm's name, is neither more nor less expensive than one that allocates it evenly.

The aggregate null effect is a cancellation rather than a set of small effects. The most notable difference in the effect of removing rank in equilibrium is between $+3.9\%$ for (B,G,G,G) and -2.6% for (G,G,G,B). These two configurations have the same J and the same number of generics; the main difference is which firm is prominent under alphabetical ranking.

The design question is therefore how to allocate prominence. Alphabetical ranking is costly only when the product it favours is expensive, but the pricing pressure runs counter to that intuition. Section 7.1 identifies the mechanism, and Section 7.2 considers allocating the position by price.

The pricing response to prominence. Removing rank can either raise or lower expenditures, and the direction depends on which firm held the prominent position. In (G,G,G,B), the first-ranked generic initially holds the prescription-loyal demand. Removing rank takes that away, and the firm lowers its price by 9.9%, from

0.332 to 0.299 in expectation, while the lower-ranked generics raise prices slightly. Expenditures fall by 2.6%. Conversely, in (B,G,G,G) prominence initially goes to the branded product. Removing rank sends prescription-loyal demand toward the generics, which all raise prices, most sharply the last-ranked firm, from 0.229 to 0.244. Expenditures rise by 3.9%.

Part of this asymmetry is an artifact of holding branded prices fixed. In (G,G,G,B) the firm that loses prominence can respond by cutting, and that cut enters the expenditure average. In (B,G,G,G) the firm that loses prominence is the branded, whose price does not adjust in our counterfactuals, so only the generics' upward adjustment is present. A branded firm that did respond to losing prominence would cut its price, making the $+3.9\%$ an upper bound on the increase.

The sign pattern holds across configurations. Where the branded product is listed first, removing the rank effect raises equilibrium expenditures: $+2.1\%$ in (B,G,G) and $+3.9\%$ in (B,G,G,G). Where it is listed last, expenditures fall: -1.9% in (G,G,B) and -2.6% in (G,G,G,B). In markets without a branded product the effects are close to zero in each case: -0.3%, -0.1% and $+0.7\%$ for two, three and four firms.

Overall, the results are consistent with prominence creating upward pricing pressure. When a firm loses prominence it lowers its price, and vice versa. This explains the null in all-generic markets: concentrating prominence on one firm or spreading it evenly redistributes market power among firms that all respond in price, with no bottom-line effect on expenditures.

The model also replicates the reduced-form price gradient of Section 4. Baseline duopoly prices fall from 0.378 at rank 1 to 0.358 at rank 2, a gradient of -5.3% per

TABLE 10. Demand Effect: Counterfactuals with Fixed Prices

	Baseline	Random Rank	Price Rank	Prescribe Cheapest
(G, G)	1.000	0.999	0.989	0.955
(B, G, G)	1.000	1.000	0.982	0.918
(G, G, B)	1.000	0.999	0.982	0.922
(G, G, G)	1.000	0.997	0.976	0.910
(G, G, G, G)	1.000	0.996	0.973	0.896
(B, G, G, G)	1.000	0.990	0.965	0.874
(G, G, G, B)	1.000	1.004	0.980	0.891
Average	1.000	0.999	0.979	0.910

Note: The table shows expected expenditure per consumer in each counterfactual, relative to the baseline, holding price distributions fixed at the model-implied baseline values. Since there is no outside option, expenditure equals the expected price. The "Average" row is weighted by the number of transactions in each market configuration.

TABLE 11. Equilibrium Effect: Counterfactuals with Equilibrium Prices

	Baseline	Random Rank	Price Rank	Prescribe Cheapest
(G, G)	1.000	0.997	0.888	0.612
(B, G, G)	1.000	1.021	0.973	0.833
(G, G, B)	1.000	0.981	0.935	0.797
(G, G, G)	1.000	0.999	0.868	0.631
(G, G, G, G)	1.000	1.007	0.892	0.726
(B, G, G, G)	1.000	1.039	0.950	0.743
(G, G, G, B)	1.000	0.974	0.900	0.711
Average	1.000	0.998	0.923	0.747

Note: The table shows equilibrium expenditures in each counterfactual, relative to the baseline. The "Average" row is weighted by the number of transactions in each market configuration. Table A.12 shows the same table without the normalization to baseline.

rank, against the -5.3% per rank estimated for duopoly markets in Table 5. While the comparison is not completely like-for-like, this is still reassuring.

7.2. Price ranking

Sorting the list by price lowers equilibrium expenditures by 7.7%, against 2.1% when prices are held fixed. The gap is the bias of ignoring the supply response: a fixed-price evaluation of this reform would understate it by a factor of 3.7. The savings are unsurprising in direction, since prominence now accrues to the cheapest firm; the point of interest is their size and where they fall.

The demand and supply channels are strongest in opposite configurations, and the contrast is informative about the size of the supply response. Steering prescriptions saves most where the prominent product is expensive, so at fixed prices the largest savings occur when the branded is first: expenditures fall to 0.965 in (B,G,G,G) against 0.980 in (G,G,G,B). Allowing prices to adjust reverses the ordering. In (B,G,G,G) expenditures fall only a little further, to 0.950, because the firms gaining demand from the branded face the upward pressure described above. In (G,G,G,B) they fall much further, to 0.900, because the reform strips prominence from a firm that responds by cutting. The supply response is thus roughly five times the demand response in the second case and a fifth of it in the first. We do not push the comparison further, since the two configurations differ in other respects, including the empirical distribution of branded prices, which is lower in (G,G,G,B) (Figure 5, panels (b) and (c)).

Holding the number of strategic firms fixed, savings are ordered by the branded product's presence and position. With two generics, price ranking saves 11.2% in (G,G), 6.5% in (G,G,B) and 2.7% in (B,G,G). With three generics, it saves 13.2% in (G,G,G), 10.0% in (G,G,G,B) and 5.0% in (B,G,G,G). The ordering replicates across both groups. Branded demand cannot be won by undercutting, so its presence shrinks the prize that price ranking makes contestable, and a first-ranked branded absorbs more prescription-loyal demand than a last-ranked one, shrinking it further. The effect of price ranking is therefore governed less by the number of competitors than by how much of the loyal demand is competed for at all: (G,G,G,G), with four strategic firms, saves 10.8%, less than (G,G,G) with three.

Price ranking makes the prominent position contestable. A minority of consumers are close to inelastic with respect to the pharmacy's substitution offer: the random coefficient on the prescription dummy has a standard deviation of $\hat{\sigma} = 1.73$, so two standard deviations above the mean the utility premium from being prescribed is $0.906 + 2 \times 1.73 = 4.37$, against a generic substitution bonus of $\hat{\beta}_2 = 1.89$. For these consumers the physician's choice determines the purchase. Under alphabetical ranking this segment accrues to whichever firm the alphabet favours, and the rent attached to it is never competed for. Under price ranking it accrues to the cheapest firm, which steepens the expected profit function in the neighbourhood of the lowest rival price and shifts the equilibrium distribution downward. The demand parameters are identical across all scenarios; no consumer becomes better informed or more price-sensitive.

7.3. Prescribe cheapest

Prescribing the cheapest product yields equilibrium savings of 25.3% on average, against 9.0% at fixed prices, with the largest effects in all-generic markets (38.8% in (G,G), 36.9% in (G,G,G)) and the smallest where a branded product is present (16.7% in (B,G,G), 20.3% in (G,G,B)). Thus perfect steering yields 3.3 times larger savings than price ranking. While this policy has been actively discussed in Danish policy circles, it is clearly more extreme in its interference with physician autonomy. It nevertheless serves as a benchmark for the maximal price effect achievable through changing physician prescribing.

These results speak to how choice architecture is evaluated in markets where firms set prices. A literature documents that defaults and information provision steer consumers toward better choices, and typically measures the effect holding prices

TABLE 12. Equilibrium Expected Prices by Rank

Market Structure	Rank	Baseline	Random Rank	Price Rank	Prescribe Cheapest
(G, G)	Rank 1	0.378	0.367	0.336	0.256
	Rank 2	0.358	0.367	0.336	0.256
(B, G, G)	Rank 2	0.425	0.430	0.412	0.370
	Rank 3	0.415	0.428	0.410	0.370
(G, G, B)	Rank 1	0.437	0.418	0.400	0.356
	Rank 2	0.413	0.417	0.399	0.358
(G, G, G)	Rank 1	0.275	0.238	0.218	0.183
	Rank 2	0.231	0.241	0.219	0.193
	Rank 3	0.212	0.235	0.216	0.191
(G, G, G, G)	Rank 1	0.225	0.176	0.167	0.171
	Rank 2	0.181	0.180	0.172	0.111
	Rank 3	0.159	0.176	0.167	0.191
	Rank 4	0.150	0.177	0.166	0.179
(B, G, G, G)	Rank 2	0.249	0.246	0.230	0.209
	Rank 3	0.236	0.244	0.229	0.210
	Rank 4	0.229	0.244	0.229	0.215
(G, G, G, B)	Rank 1	0.332	0.299	0.283	0.257
	Rank 2	0.297	0.298	0.283	0.261
	Rank 3	0.286	0.299	0.283	0.264

Note: The table shows the expected price at each generic list position under each counterfactual. Prices are expressed as a fraction of the pre-patent-expiration price of the original manufacturer. Branded firms' prices are held fixed at the empirical distribution and are not shown. Under alphabetical ranking a firm's position is fixed by its name; under price ranking and prescribe cheapest, position is determined by the realized price and is therefore an outcome, so a position should not be read as following the same firm across columns.

fixed.[32] In our setting the supply response is roughly three times the demand-side effect, so a fixed-price estimate is a lower bound on the value of the reform.

7.4. Limitations

Four limitations of the counterfactual exercise deserve mention.

The first is coverage. We solve configurations with $J \leq 4$ and at most one branded product. Markets with at most 4 firms cover 52.7% of transactions and 59.7% of

32. See e.g. Abaluck and Gruber (2011); Madrian and Shea (2001); Bhargava et al. (2017); Ketcham et al. (2019).

product-periods. However, the reduced form results from Table 5 showed that the rank effect diminished as the number of firms went to 8. We thus view our results as perhaps an upper bound if one extrapolated to more competitive markets.

The second is that market structure is held fixed, which may change the welfare accounting (Ryan, 2012). A reduction in margins of 7.7% could reduce the number of firms, at least partly undoing the original savings effect. However, since the products are extremely close substitutes compared to many other goods markets, a reduction in entry may also reduce duplicated fixed entry costs (Mankiw and Whinston, 1986).

The third is product availability. Our demand model includes an indicator for delivery failure (Table 7), which we hold fixed across counterfactuals. Thus, our counterfactual experiments do not capture pricing responses under alternative architectures. Downward pressure on prices has been linked to shortages in the United States (Yurukoglu et al., 2017), but Kortelainen et al. (2023) find that availability was not adversely affected by reforms in the Nordics even though expenditures were successfully reduced.

Finally, we cannot explicitly rule out the possibility of collusion in our data. And there is empirical evidence documenting dynamic price patterns consistent with cycles (Hauschultz and Munk-Nielsen, 2017; Janssen, 2022). However, we note that the price-rank gradient of our model in the baseline aligns almost exaclty with those we estimate from the data, which indicates that the competitive model agrees with what happens in the data.

8. Conclusion

We estimate the effect of product prominence in Danish pharmaceutical markets, using alphabetical ordering in physicians' prescribing software as variation in prominence that no firm controls. Products listed earlier are prescribed more often, prescriptions carry through to purchases despite a pharmacist obliged to offer the cheapest alternative, and prices fall with alphabetical position. Prominence has been shown to shift quantities in many settings; documenting that it shifts prices requires variation in which product is prominent that is unrelated to the firm's own conduct, which alphabetical ordering provides. Firms respond to where they sit on the list, which is what makes reordering it a policy instrument at all.

The allocation of prominence affects demand and supply differently. Steering prescriptions toward the cheapest product lowers expenditures, and particularly so when the displaced product is expensive. But prominence also confers inelastic demand: our estimates imply substantial heterogeneity in how closely patients follow the prescription. A firm that is handed the prescription-loyal segment exogenously through prominence raises its price. This is why allocating prominence on name is neither better nor worse than distributing it evenly, it is simply a redistribution of market power. What matters is making firms compete for prominence, and sorting the list by price achieves this. And the greatest competitive response happens in the markets with only generic firms, because the branded tends to command its own loyal segment.

Where the demand is contestable, most of the effect comes from the supply side. Sorting the list by price reduces equilibrium expenditures by 7.7%, but only 2.1%-points if prices are held fixed. The implication reaches beyond pharmaceuticals: a large literature evaluates defaults and information provision holding prices fixed. Our findings indicate that this may only be a minor part of the total effect. The results also have direct policy implications: removing frictions accomplishes little beyond arbitrary redistribution across firms. While generic prescribing would achieve a far greater total reduction in expenditures of 25.3%, that is possibly a politically sensitive policy interfering more with physician autonomy. Reordering the list restricts no one, and works by changing what firms stand to gain from undercutting. The same margin exists wherever an institution directs demand toward whichever seller is cheapest.

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Appendix Intended for Online Publication

TABLE A.1. Sample Selection

Step	Criterion	Product-periods
	Raw data	3,701,975
A	After April 2005	2,162,861
B	Off patent $\geq$ 1 year	1,245,500
C	Company name observed	1,000,414
D	# firms ≥ 2	832,585
E	# firms ≤ 8	783,872
F	Price, quantity and prescriptions observed	749,859
G	Only generics (quotes in name)	359,575
H	# branded $\leq \min(2, J-2)$	293,075

Note: The sample "F" is used for our primary reduced form results and the basis for our further sample restrictions in the micro sample, see Table A.2. The sample "H" is used for our robustness checks using generic products only, see Table A.8, and the conditions must be satisfied for at least one product in our micro sample (Table A.2).

TABLE A.2. Construction of the micro sample

Step	Restriction	Dropped	Observations
0	Transactions sample		232,868,657
1	Random subsample	98.71%	3,000,000
2	Some product satisfies G & H	54.77%	1,356,817
3	No missing product data in market	6.47%	1,269,002
4	Prescribed in another market	3.41%	1,225,686
5	Missing demographics	18.01%	1,004,956
6	Substitution prohibited	3.68%	967,930
7	Subsidy information missing	0.00%	967,914

Note: The table starts from 3 million randomly selected transactions from our full set of permissible transactions. Step 2 requires that at least one product must satisfy criterion G and H in Table A.1. Note that this imposes a minimum requirement on how many generics we observe, for instance ruling out all-branded markets. Step 3 rules out missing values for *any* product in the market (including ones that are not chosen). Step 4 rules out prescriptions out-of-market, which can only ever occur if for example the pharmacist takes one prescription for a 100-pill package and splits it into two 50-pill packages — then we cannot assign the prescription dummy. Step 6 rules out that the physician has marked the "no substitution" option; this option is rarely used and its use over time changes dramatically, so we have opted to drop those observations to avoid estimate contamination. The final sample is used for estimation of the structural search model in Section 5.

TABLE A.3. Summary Statistics – Transaction Sample

	Mean	Std.	P10	P50	P90
Income (log)	11.90	1.22	11.41	12.00	12.60
Age	60.18	17.11	36.00	63.00	81.00
1(Female)	0.55	0.50	0.00	1.00	1.00
1(Education post 2nd)	0.53	0.50	0.00	1.00	1.00
No. products	5.19	1.66	3.00	5.00	7.00
No. generic	4.10	1.70	2.00	4.00	6.00
1(Any delivery failure)	0.30	0.46	0.00	0.00	1.00
Share of delivery failures	0.07	0.13	0.00	0.00	0.25
1(Winner prescribed)	0.24	0.43	0.00	0.00	1.00
1(Generic prescribed)	0.65	0.48	0.00	1.00	1.00
1(Parallel import prescribed)	0.02	0.12	0.00	0.00	0.00
Alphabetical rank prescribed	2.52	1.74	1.00	2.00	5.00
1(Winner purchased)	0.64	0.48	0.00	1.00	1.00
1(Generic purchased)	0.85	0.36	0.00	1.00	1.00
1(Parallel import purchased)	0.02	0.13	0.00	0.00	0.00
Alphabetical rank purchased	2.86	1.60	1.00	3.00	5.00
Observations (transactions)					967,914

Note: This sample is used for estimation of the structural search model in Section 5.

TABLE A.4. Markov Transitions for Winner Status

		Winner$_{t+1}$ 0	Winner$_{t+1}$ 1
Winner$_t$	0	84.7%	15.3%
	1	27.9%	72.1%

Note: Winner Status is equal to 1 if the product has the minimum price in the market during that period (so multiple products can win if there is a tie). The table is based on all observations of products where two subsequent periods were observed for the same product (741,372 product-periods).

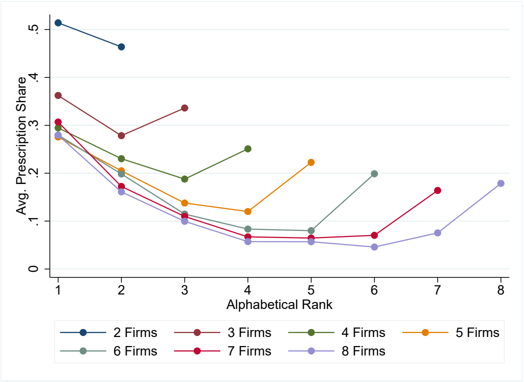
Appendix A: Additional Results

A.1. Robustness: Only Generic Products

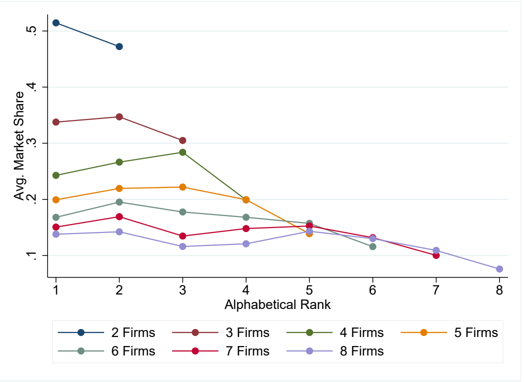
Table A.8 shows the results of our main regressions, but estimated on the subsample "H" in Table A.1, which is only generics products, and excludes markets with fewer than two generics.

FIGURE A.1. Alphabetical Rank and Market Outcomes: All Products

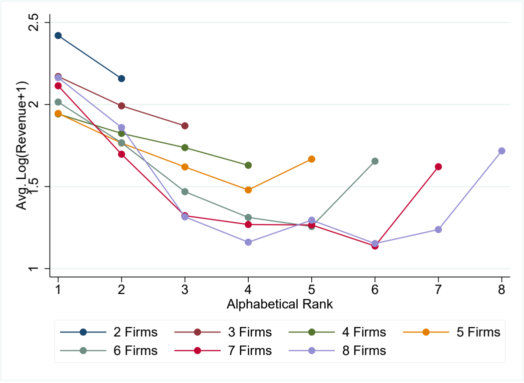
(A) Prescription share



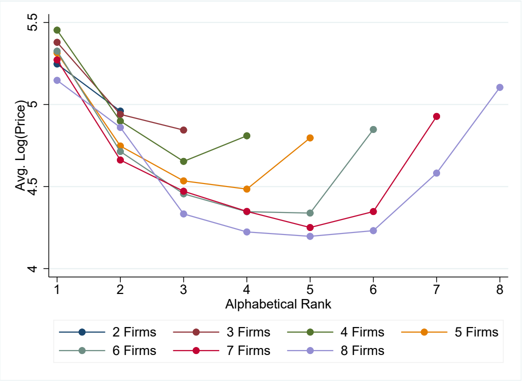
(B) Market share



(C) Revenue



(D) Price



Note: The figure is constructed using our product-level dataset (i.e. an observation is a product-period) including all products both branded and generic, unlike Figure 2, which only shows generics. Since 5% of observations have zero revenue we add 1 to revenue (in 1000 DKK) before taking logs.

A.2. Robustness: Branded First or Last Alphabetically

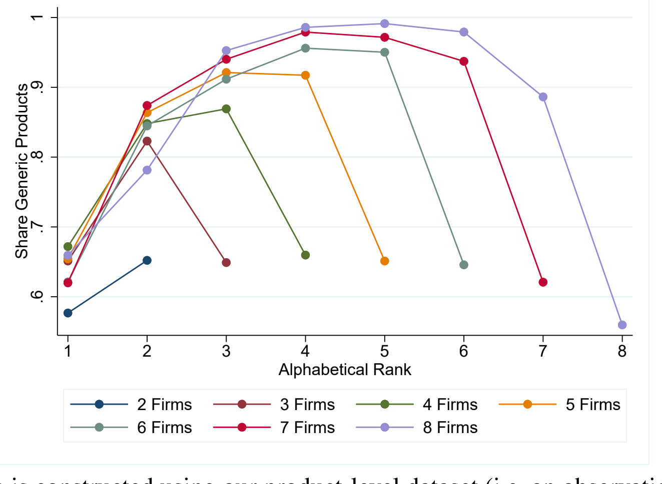
Table A.9 explores if our results are robust if we estimate on the subsample where the generics are before or after the branded product(s), using a time-varying classification of whether the first-ranked product is generic. Below, we describe this classification in detail, along with an alternative time-invariant, ATC-level classification.

TABLE A.5. Markov Transitions for Alphabetical Rank

		Rank$_{t+1}$							
		1	2	3	4	5	6	7	8
Rank$_t$	1	99.4	0.6	0.0	0.0	0.0	0.0	0.0	0.0
	2	1.0	97.4	1.6	0.1	0.0	0.0	0.0	0.0
	3	0.1	3.0	94.2	2.6	0.1	0.0	0.0	0.0
	4	0.1	0.3	4.5	91.9	3.2	0.1	0.0	0.0
	5	0.0	0.1	0.4	6.3	89.6	3.6	0.1	0.0
	6	0.0	0.0	0.1	0.6	7.8	87.6	3.8	0.1
	7	0.0	0.0	0.1	0.2	0.9	9.3	86.6	2.9
	8	0.0	0.0	0.0	0.2	0.3	1.4	11.7	86.4
	Obs.	291,488	190,774	100,672	71,660	42,560	22,543	10,597	3,087

Note: Table based on all observations of products where two subsequent periods were observed (741,372 product-periods).

FIGURE A.2. Generic share (Wide definition) by rank



Note: The figure is constructed using our product-level dataset (i.e. an observation is a product-period) including all products both branded and generic. Note that there can be multiple branded in a given market, although there is most typically only one.

The time-varying classification: Here we just check if the first-ranked product is generic (has quotes in the name) or not. Our sample of 749,859 product-periods fall on 230,561 market-periods, of which 31.9% have a generic ranked first.

The time-invariant classification: In our market-level dataset, we have 387 unique ATC codes. Of these, 61 have the branded first, 53 have the generic first, and 131 have

TABLE A.6. Tabulation of Entry and Exit Events by Rank and Number of Firms

	Number of firms							
	2	3	4	5	6	7	8	Total
Panel A: Exit events								
1	426	347	246	213	169	62	40	1503
2	258	272	218	195	134	106	51	1234
3		141	165	205	135	85	47	778
4			135	140	124	72	44	515
5				94	75	62	35	266
6					49	62	40	151
7						38	32	70
8							26	26
Total	684	760	764	847	686	487	315	4543
Panel B: Entry events								
1	439	372	217	129	93	45	22	1317
2	209	227	214	165	94	59	31	999
3		94	139	124	84	65	38	544
4			53	92	73	51	33	302
5				61	71	41	24	197
6					23	44	34	101
7						26	24	50
8							13	13
Total	648	693	623	571	438	331	219	3523

Note: The table takes our main sample of 749,859 observations and counts each entry or exit event for these. An entry is the first observation of a product, and an exit is the last, except if that period is the first/last of the sample. The columns indicate the number of firms in the market: for entry events it is after entry, and for exit events it is before exit. And the rows indicates the rank of the product in question where it enters to or exits from. Note that there are also some products, which have been excluded by our sample selection criteria, but which still affect the ranks of incumbent products in our sample. Thus, there are more exit/entry events than those shown above which contribute to the variation in alphabetical rank in our regressions.

both occurring at different points in time. The remaining 142 ATC codes are either missing the branded or generic product, or we were not able to find the name of the branded and generic molecules. There are several substitution groups (markets) per ATC code, and each inherits the first/last status from the ATC code. In the robustness check, we use the ATC codes that have either branded or generic first, meaning that we can use only 29.5% of ATC codes.

TABLE A.7. Purchase Conditional on Prescription

	Prescription subsample						
Purchased product	Full	Rank == 1	A	B	C	Generic	Parallel I.
Prescribed	0.4529	0.4895	0.7819	0.4912	0.2449	0.4100	0.3573
A	0.6489	0.6632	0.9001	0.4215	0.5605	0.6645	0.6725
B	0.1465	0.1566	0.0530	0.5539	0.0765	0.1282	0.2040
C	0.2046	0.1802	0.0469	0.0246	0.3630	0.2073	0.1235
Rank	2.4624	1.9171	2.0307	2.1668	2.8253	2.8844	1.4829
Rank == 1	0.3523	0.5976	0.4487	0.4231	0.2701	0.2389	0.6946
Generic	0.4928	0.4415	0.3821	0.4064	0.5881	0.9109	0.1613
Parallel I.	0.0775	0.1222	0.0662	0.1202	0.0710	0.0185	0.6504
Observations	2328637	965087	747649	374668	1167820	808189	129460

Note: Each column is a sample defined by some feature of the *prescribed* product. The rows are then summary statistics for the variables indicated, which are characteristics of the *purchased* product. The sample are all transactions where the purchased product is in our market-level sample, “F” in Table A.1.

A.3. Robustness to Entry and Exit of Products

Table A.11 shows the results of our main regressions when we restrict the sample to products that are present in all periods (i.e. we drop products that enter or exit during the sample period). Products that are present before the first period (April 2005) or after the last period (June 2017) are kept in the sample. An entry is defined as the first period where a product is present in a market, so long as that period is after the first observed period for the market. Similarly, exit is defined as the last period where a product is present in a market, so long as that period is before the last observed period for the market.

TABLE A.8. Sample: Only Generic Products

	(1)	(2)	(3)	(4)	(5)
Panel A: Outcome = Prescription share					
Alphabetical Rank	-0.136***	-0.113***	-0.157***	-0.116***	-0.178***
	(0.00958)	(0.00830)	(0.00859)	(0.00733)	(0.00997)
Alphabetical Rank × No. Firms	0.0137***	0.0124***	0.0174***	0.0136***	0.0199***
	(0.00143)	(0.00121)	(0.00121)	(0.000929)	(0.00136)
Panel B: Outcome = Market share					
Alphabetical Rank	-0.0215*	-0.0265***	-0.0224**	-0.00822	-0.0197*
	(0.00874)	(0.00699)	(0.00802)	(0.00769)	(0.00936)
Alphabetical Rank × No. Firms	0.00148	0.00213*	0.00152	-0.000613	0.00142
	(0.00133)	(0.00106)	(0.00114)	(0.00104)	(0.00127)
Panel C: Outcome = log(Price)					
Alphabetical Rank	-0.246***	-0.204***	-0.0367*	-0.0486*	-0.0400**
	(0.0419)	(0.0480)	(0.0156)	(0.0221)	(0.0121)
Alphabetical Rank × No. Firms	0.0389***	0.0284***	0.00727**	0.00856**	0.00800***
	(0.00660)	(0.00622)	(0.00226)	(0.00269)	(0.00186)
Panel D: Outcome = Revenue share					
Alphabetical Rank	-0.0243**	-0.0261***	-0.0249***	-0.00746	-0.0246**
	(0.00832)	(0.00652)	(0.00734)	(0.00685)	(0.00860)
Alphabetical Rank × No. Firms	0.00222	0.00240*	0.00225*	-0.000166	0.00232*
	(0.00125)	(0.000964)	(0.00103)	(0.000883)	(0.00115)
Drug age FE	Yes	Yes	Yes	Yes	No
No. Firms FE	Yes	Yes	Yes	Yes	No
Company FE	No	Yes	Yes	No	Yes
Period FE	No	Yes	Yes	Yes	No
Market FE	No	No	Yes	No	No
Product FE	No	No	No	Yes	No
Market-by-date FE	No	No	No	No	Yes
Clusters	698	698	698	695	698
Observations	293075	293073	293073	293002	293073

Note: Standard errors clustered at the market level, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$. Sample: “H” from Table A.1: only generics, and only markets with at least two generics.

TABLE A.9. Robustness: generics first or last (Outcome: log(price))

	(1) Full	(2) OM first	(3) Generic first
Alphabetical Rank	-0.0696***	0.00530	-0.133***
	(0.0168)	(0.0211)	(0.0266)
Alphabetical Rank×No. Firms	0.00845***	-0.00378	0.0220***
	(0.00199)	(0.00250)	(0.00323)
Drug age FE	Yes	Yes	Yes
No. Firms FE	Yes	Yes	Yes
Period FE	Yes	Yes	Yes
Product FE	Yes	Yes	Yes
Clusters	1563	1236	578
Observations	749741	487985	261633

Note: First/last classification is *period*-based, and thus time-varying. That is, for each period, we check if the first-ranked product is branded or generic. Standard errors clustered at the market level, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$. In columns (2) and (3), the sample is split based on whether the branded name or the generic name comes first alphabetically. The outcome variable is the logarithm of the retail price per DDD.

TABLE A.10. Robustness: generics first – All specifications

	(1)	(2)	(3)	(4)	(5)
Panel A: Outcome = Prescription share					
Alphabetical Rank	0.0327*	-0.0316**	-0.0546***	-0.0826***	-0.0608***
	(0.0166)	(0.0116)	(0.0121)	(0.00996)	(0.0128)
Alphabetical Rank×No. Firms	-0.00655**	0.00252	0.00581***	0.0106***	0.00681***
	(0.00249)	(0.00170)	(0.00169)	(0.00132)	(0.00172)
Panel B: Outcome = Market share					
Alphabetical Rank	-0.0677***	-0.0492***	-0.0393***	-0.0365***	-0.0398***
	(0.0101)	(0.00812)	(0.00809)	(0.00858)	(0.00858)
Alphabetical Rank×No. Firms	0.00702***	0.00507***	0.00435***	0.00395***	0.00449***
	(0.00154)	(0.00119)	(0.00116)	(0.00112)	(0.00119)
Panel C: Outcome = log(Price)					
Alphabetical Rank	0.0294	-0.121*	-0.0592*	-0.133***	-0.0357
	(0.0239)	(0.0475)	(0.0237)	(0.0266)	(0.0242)
Alphabetical Rank×No. Firms	0.0207***	0.0327***	0.0252***	0.0220***	0.0238***
	(0.00418)	(0.00576)	(0.00358)	(0.00323)	(0.00359)
Panel D: Outcome = Revenue share					
Alphabetical Rank	-0.0449***	-0.0399***	-0.0340***	-0.0345***	-0.0352***
	(0.0101)	(0.00803)	(0.00817)	(0.00780)	(0.00866)
Alphabetical Rank×No. Firms	0.00749***	0.00697***	0.00647***	0.00617***	0.00674***
	(0.00157)	(0.00121)	(0.00119)	(0.00111)	(0.00121)
Drug age FE	Yes	Yes	Yes	Yes	No
No. Firms FE	Yes	Yes	Yes	Yes	No
Company FE	No	Yes	Yes	No	Yes
Period FE	No	Yes	Yes	Yes	No
Market FE	No	No	Yes	No	No
Product FE	No	No	No	Yes	No
Market-by-date FE	No	No	No	No	Yes
Clusters	608	608	606	578	593
Observations	261770	261768	261766	261633	260357

Note: Subsample where the branded product is always placed last. This uses the time-varying classification of first/last, same as in Table A.9. Standard errors clustered at the market level.

TABLE A.11. Robustness: Dropping Products that Enter or Exit

	(1)	(2)	(3)	(4)	(5)
Panel A — Outcome: Prescription share					
Alphabetical Rank	-0.0464**	-0.0439***	-0.0394**	-0.0598***	-0.0319
	(0.0146)	(0.0114)	(0.0131)	(0.0118)	(0.0174)
Alphabetical Rank×No. Firms	0.00216	0.00365*	0.00295	0.00560***	0.00184
	(0.00222)	(0.00162)	(0.00174)	(0.00130)	(0.00238)
Panel B — Outcome: Market share					
Alphabetical Rank	-0.0210*	-0.0340***	-0.00118	-0.0260**	0.0151
	(0.0101)	(0.00933)	(0.0120)	(0.00929)	(0.0159)
Alphabetical Rank×No. Firms	0.00252	0.00339*	-0.000652	0.00111	-0.00277
	(0.00151)	(0.00136)	(0.00160)	(0.00117)	(0.00216)
Panel C — Outcome: log(Price)					
Alphabetical Rank	-0.649***	-0.408***	-0.0914**	-0.0439	-0.105*
	(0.0895)	(0.0738)	(0.0338)	(0.0254)	(0.0443)
Alphabetical Rank×No. Firms	0.0801***	0.0538***	0.0105*	0.00442	0.0118
	(0.0137)	(0.0110)	(0.00471)	(0.00320)	(0.00622)
Panel D — Outcome: Revenue share					
Alphabetical Rank	-0.0161	-0.0199*	-0.00532	-0.0154	0.00918
	(0.0101)	(0.00933)	(0.0113)	(0.00962)	(0.0149)
Alphabetical Rank×No. Firms	0.000604	0.00118	-0.000198	0.000967	-0.00255
	(0.00163)	(0.00144)	(0.00163)	(0.00132)	(0.00217)
Drug age FE	Yes	Yes	Yes	Yes	No
No. Firms FE	Yes	Yes	Yes	Yes	No
Company FE	No	Yes	Yes	No	Yes
Period FE	No	Yes	Yes	Yes	No
Market FE	No	No	Yes	No	No
Product FE	No	No	No	Yes	No
Market-by-date FE	No	No	No	No	Yes
Clusters	1297	1297	1286	1287	501
Observations	281463	281462	281451	281419	163031

Note: Standard errors clustered at the market level, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$. The sample drops products that enter or exit during the sample window.

TABLE A.12. Equilibrium Effect: Counterfactuals with Equilibrium Prices — Levels

	Baseline	Random Rank	Price Rank	Prescribe Cheapest
(G, G)	0.302	0.301	0.268	0.185
(B, G, G)	0.240	0.245	0.233	0.200
(G, G, B)	0.262	0.257	0.244	0.209
(G, G, G)	0.121	0.121	0.105	0.076
(G, G, G, G)	0.052	0.052	0.046	0.038
(B, G, G, G)	0.113	0.117	0.107	0.084
(G, G, G, B)	0.158	0.154	0.142	0.112
Average	0.193	0.192	0.179	0.146

Note: The table shows the same results as in Table 11, but without normalizing by the baseline. The "Average" row is weighted by the number of transactions in each market configuration. Prices are normalized by the pre-patent price. Since marginal cost is estimated at zero, a value of one is the pre-patent per-period profit of a monopolist.

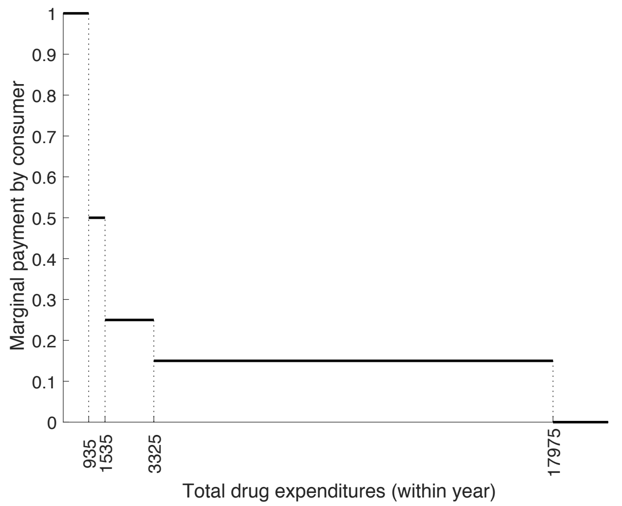
Appendix B: Institutional Details

This section covers additional institutional details for completeness.

Insurance: As mentioned in Section 2.1, all Danes are covered by public health insurance, which covers a part of the costs for the cheapest drug in the market. Figure B.1 shows the marginal co-payment rate for the consumer as a function of the total expenditures within a year. Each consumer's year begins at the first purchase and then lasts until the same date of the following year, so they are not synchronized across consumers.

Pharmacy regulation: Pharmacist margins on prescriptions are dictated by the government and do not depend on consumer choice. The logistics of transporting products to pharmacies is carried out by a fully regulated state-mandated duopoly that is independent of the pharmacies. Pharmacies are required to stock the winning product, and can typically obtain any product within a few days. It is technically legal to buy and sell prescription drugs outside this system, but since the government subsidy will not apply in that case, no pharmacy does this. The pharmacy industry as a whole is heavily regulated with respect to entry, margins and ownership structure (pharmacies must be owned by a pharmacist, which for instance bars supermarket chains from entering the market). Additionally, pharmacies have an internal reallocation method whereby high revenue pharmacies, e.g. in high-demand areas like urban centres, have to pay a part of their revenue to those with lower revenues. In 2015, entry was deregulated, allowing free entry and deregulating ownership structure. Subsequently, the number of pharmacies greatly increased. Apart from lower waiting times at pharmacies, we do not expect that this greatly affected demand for prescription pharmaceuticals.

FIGURE B.1. Out of pocket payment under the public health insurance



Note: The plot depicts the fraction of the transaction price that must be paid out of pocket by the consumer. The spending thresholds are as of 2015 and subject to annual inflation adjustments. Each consumer will have an asynchronous drug-expenditure year: the first time a consumer makes a purchase, an expenditure year is initiated. During the year, expenditures mount and the marginal payment starts to fall. Then, precisely a year later, total expenditures are reset to zero. The next year does not begin until the next time the consumer makes a purchase.

Detailing. In Denmark, prescription drugs may be promoted directly to health professionals, including through medical sales representatives, although advertising of prescription drugs to the general public is prohibited and physician-directed promotion is regulated (Danish Ministry of Health, 2022). Detailing was historically non-negligible: Schramm et al. (2007) recorded 1,050 representative visits to 47 Danish GPs over six months.

Importantly for our setting, however, the clearest Danish evidence on the effect of detailing concerns a broader therapeutic margin than our markets. Søndergaard et al. (2009) study 832 representative visits promoting Symbicort (budesonide/formoterol) and find that the first visit substantially increased its share among fixed ICS/LABA combinations (odds ratio 2.39), while having essentially no effect on overall steroid treatment. Thus, the documented margin is substitution across therapeutic combinations rather than between bioequivalent manufacturers within a substitution group. Our markets are much narrower, comprising products with the same substance, strength and dose, and are observed only after generic entry. Pharmacy-level generic substitution further reduces the incentive to promote a particular manufacturer within such a market. Consistent with this mechanism, Timonen et al. (2010) find that following the introduction of generic substitution in Finland, generic firms increased representative visits to pharmacies, while physician visits generally did not increase. We therefore view detailing as a less likely source of within-market variation in our setting, while now discussing it explicitly as a potential demand shifter, particularly for branded products. Moreover, persistent product-specific differences in detailing intensity are absorbed by the product fixed effects in our reduced form specification.

Appendix C: Why the Standard Markup Inversion Fails

The standard approach to marginal cost assumes the observed prices are a Nash equilibrium and inverts each firm's first-order condition. The condition, however, describes only the smooth part of the firm's problem. This appendix shows that the cost it returns is one at which the firm would rather undercut the market leader than charge the price it was observed charging, and that no profile with a tie at the lowest price is an equilibrium for any costs at all.

C.1. Setup

Consider $J \geq 2$ firms competing in prices, with a single representative consumer whose indirect utility from product j is $\delta_j + \alpha \log p_j + \beta \mathbf{1}\{p_j = \min_k p_k\} + \varepsilon_j$, where $\alpha < 0$,

$|\alpha| \equiv -\alpha$, and $\beta \geq 0$ is the prominence bonus accruing to the cheapest product. This simplifies the demand system of Section 5 to one consumer type and a single step at the minimum price, retaining the two features that matter: a constant price elasticity and a discrete reward for being cheapest.

With ε type-I extreme value, and writing $\exp(\alpha \log p_j) = p_j^{\alpha}$, shares take the form

$$s_j(p) \;=\; \frac{w_j(p)}{\Lambda(p)}, \qquad w_j(p) \;=\; e^{\delta_j}\, p_j^{\alpha}\, e^{\beta \mathbf{1}\{p_j = \min_k p_k\}}, \qquad \Lambda(p) \;=\; \sum_k w_k(p). \quad \text{(C.1)}$$

Because $\log w_j$ is linear in $\log p_j$, the own-price elasticity is

$$\frac{\partial \log s_j}{\partial \log p_j} \;=\; \frac{\partial \log w_j}{\partial \log p_j} - \frac{\partial \log \Lambda}{\partial \log p_j} \;=\; \alpha - \alpha s_j \;=\; \alpha(1 - s_j). \qquad \text{(C.2)}$$

Firm j earns $\pi_j(p) = (p_j - c_j)s_j(p)$.

C.2. The inverted cost invites an undercut

Fix a profile p in which firm 1 is uniquely cheapest. For $k \neq 1$ a small change in p_k leaves the identity of the cheapest firm unchanged, so on that part of the problem the first-order condition holds and, by (C.2), delivers the Lerner form

$$\frac{p_k - c_k}{p_k} \;=\; \frac{1}{|\alpha|\,(1 - s_k)}. \qquad \text{(C.3)}$$

This is the equation the standard approach inverts for c_k, and from here on c_k is that recovered number.[33]

33. It returns a nonnegative cost only if $|\alpha|(1 - s_k) \geq 1$, otherwise the cost inversion is undefined.

Now let firm k undercut the leader, setting $\tilde{p}_k = p_1 - \eta$ and taking $\eta \downarrow 0$. Denoting post-deviation objects by tildes, the comparison to be made is

$$\frac{\tilde{\pi}_k}{\pi_k} = \frac{(p_1 - c_k)\,\tilde{s}_k}{(p_k - c_k)\,s_k} = \underbrace{\frac{p_1 - c_k}{p_k - c_k}}_{\text{margin ratio}} \times \underbrace{\frac{\tilde{s}_k}{s_k}}_{\text{share ratio}} . \tag{C.4}$$

The margin ratio is below one, since the firm cuts its price at unchanged cost; the share ratio is above one. We evaluate the first exactly and bound the second from below.

PROPOSITION C.1. *Let firm 1 be uniquely cheapest and let c_k solve (C.3). Firm k strictly gains from undercutting if*

$$\underbrace{\left[1 - |\alpha|(1 - s_k)\,\frac{p_k - p_1}{p_k}\right]}_{\textit{margin ratio, exactly}} \cdot \underbrace{\frac{\Gamma}{1 + (\Gamma - 1)\,s_k}}_{\textit{lower bound on share ratio}} > 1, \qquad \Gamma \equiv \left(\frac{p_k}{p_1}\right)^{|\alpha|} e^{\beta}. \tag{C.5}$$

Proof. Take the two factors of (C.4) in turn.

Margin ratio. From (C.3), $p_k - c_k = p_k / \big(|\alpha|(1 - s_k)\big)$. Writing $p_1 - c_k = (p_k - c_k) - (p_k - p_1)$ and dividing by $p_k - c_k$,

$$\frac{p_1 - c_k}{p_k - c_k} = 1 - \frac{p_k - p_1}{p_k - c_k} = 1 - |\alpha|(1 - s_k)\,\frac{p_k - p_1}{p_k},$$

which is the first bracket of (C.5), with equality.

Share ratio. The deviation changes exactly two weights in (C.1): w_1 and w_k. Firm k lowers its price and gains the bonus, so $\tilde{w}_k / w_k = (p_1/p_k)^{\alpha} e^{\beta} = (p_k/p_1)^{|\alpha|} e^{\beta} = \Gamma > 1$, where the exponent flips sign because $\alpha < 0$. Firm 1 loses the bonus, so $\tilde{w}_1 / w_1 = e^{-\beta} < 1$. All other weights are unchanged. Hence

$$\tilde{\Lambda} = \Lambda + \underbrace{(\Gamma - 1) w_k}_{k\text{'s gain}} - \underbrace{(1 - e^{-\beta}) w_1}_{\text{the leader's loss}} \leq \Lambda + (\Gamma - 1) w_k,$$

and dividing by Λ, $\tilde{\Lambda}/\Lambda \leq 1+(\Gamma-1)s_k$. Since $\tilde{s}_k/s_k = \Gamma \cdot \Lambda/\tilde{\Lambda}$, this gives the second bracket. Multiplying the two factors yields the claim. $\square$

The inequality dropped in the second step discards the leader's loss of the bonus, which also raises firm k's share. What remains is the share firm k would obtain if its own weight rose by Γ and nothing else moved. The condition is therefore sufficient rather than necessary, and materially so in markets with few firms, where the leader's weight is a large part of the logit denominator, Λ. It buys a bound that involves only exponentiated utilities, with no need to track the denominator.

C.3. A threshold in the price gap alone

We can obtain a looser but more intuitive bound by setting Γ to its own lower bound, $\exp(\beta)$. This includes only the prominence bonus and discards the continuous share gain from the price cut and, as it happens, removes s_k from (C.5) entirely.

COROLLARY C.1. *Undercutting is strictly profitable whenever*

$$\frac{p_k - p_1}{p_k} < \bar{g} \equiv \frac{1-e^{-\beta}}{|\alpha|}. \tag{C.6}$$

Proof. The share factor in (C.5) is increasing in Γ, with derivative $(1-s_k)/\big(1+(\Gamma-1)s_k\big)^2 > 0$, and $\Gamma \geq e^{\beta}$ since $p_k > p_1$. At $\Gamma = e^{\beta}$ the factor is $e^{\beta}/\big(1+(e^{\beta}-1)s_k\big)$, whose reciprocal is $1-(1-e^{-\beta})(1-s_k)$. Condition (C.5) then reads $(1-e^{-\beta})(1-s_k) > |\alpha|(1-s_k)(p_k-p_1)/p_k$; divide by $1-s_k > 0$. $\square$

Since $1-e^{-\beta} < 1$, the threshold satisfies $\bar{g} < 1/|\alpha| = 0.57$ and rises toward that ceiling as β grows. Note that $1/|\alpha|$ is the smallest Lerner index that (C.3) can return, so no prominence bonus, however large, justifies a discount exceeding the firm's entire markup. The threshold also falls in $|\alpha|$, since more elastic demand implies a thinner markup and less room to cut price while still covering cost. Evaluating $\bar{g}$ at our estimates, with $|\alpha| = 1.75$ from Table 7 at the mean log income of Table A.3 and $\beta = 1.89$ the gain in search costs from entering the $\mathscr{A}$ region from the reference region $\mathscr{C}$, gives $\bar{g} = 0.486$. In conclusion:

$$\text{If } p_k \leq 1.94\, p_1, \text{ then undercutting is strictly profitable.}$$

In our data, the second-cheapest product satisfies this in 97.6 percent of our 230,561 market-periods. The condition is sufficient, not necessary, so the region in which the inversion fails is wider still.

C.4. Ties are never equilibria

Proposition C.1 conditions on the cost the inversion returns. The following does not, and covers every symmetric profile as a special case.

PROPOSITION C.2. *No profile in which two or more firms tie at the lowest price is a Nash equilibrium, for any J, any* $|\alpha|$*, any* $\beta > 0$ *and any costs.*

Proof. Let T, with $|T| = m \geq 2$, be the set of firms tied at the minimum price p, and take $k \in T$. If $p \leq c_k$ then firm k earns non-positive profit and any $p' > c_k$ is a strict improvement, so suppose $p > c_k$ and let k deviate to $p - \eta$. As $\eta \downarrow 0$ firm k's weight is unchanged, since it retains the bonus at an unchanged price, while each of the other $m-1$ members of T loses the bonus. Hence $\tilde{\Lambda} \to \Lambda - (1-e^{-\beta}) \sum_{j \in T \setminus \{k\}} w_j < \Lambda$, so

$\tilde{s}_k$ exceeds s_k by an amount bounded away from zero while the margin falls only by η. Profit strictly rises for η small enough. $\square$

A symmetric profile is the case $T = \{1,\ldots,J\}$: all firms tie, all share the bonus, and the deviator takes it alone. The argument uses neither the first-order condition nor any restriction on demand parameters; it is the tie itself that is unstable.

Together the two results leave nothing for an iterative procedure to converge to. Ties are ruled out by Proposition C.2; profiles with a unique leader and a small gap are ruled out by Corollary C.1; and an undercut produces a profile in which the displaced leader sits an arbitrarily small distance above the new minimum, so that (C.6) holds again. The ratchet halts only once margins are exhausted, at which point a firm prefers to abandon prominence and price against its loyal demand. This alternation is the structure of Edgeworth cycles (Maskin and Tirole, 1988) and of the clearinghouse models of Varian (1980), and it is why the equilibrium object in Section 6 is a distribution over prices rather than a vector of them.

C.5. Scope

The results concern the costs the inversion itself returns rather than arbitrary costs: they show that the standard recipe contradicts itself, which is what rules out its use here. Corollary C.1 is also a property of the logit. If one firm faced a segment of consumers with $\alpha \approx 0$ who would buy it at any price, its profit would be increasing in price over the relevant range and it would never undercut, however large the bonus. While there are heterogeneous coefficients in our model, no segment is close to price-inelastic.